

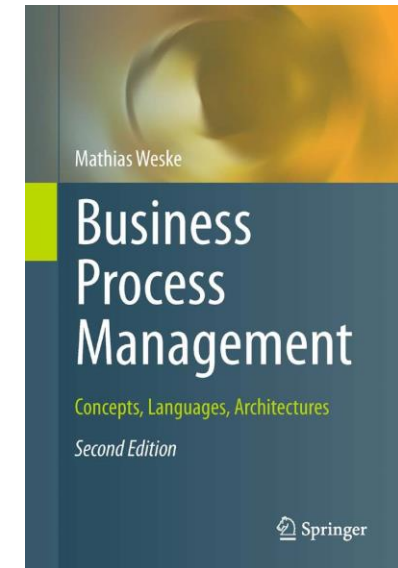
BUSINESS PROCESS MANAGEMENT

4 - BPMN

C. Salz
Feb 2025

Evolution of Enterprise System Architectures

- This PowerPoint file contains figures used in
 - Mathias Weske, Business Process Management: Concepts, Languages, Architectures 2012, XIV, 403 p. 300 illus., Hardcover ISBN: 978-3-642-28615-5, eBook ISBN: 978-3-642-28616-2 © Springer-Verlag Berlin Heidelberg 2012, 2007
- This module is based on that book



Starter

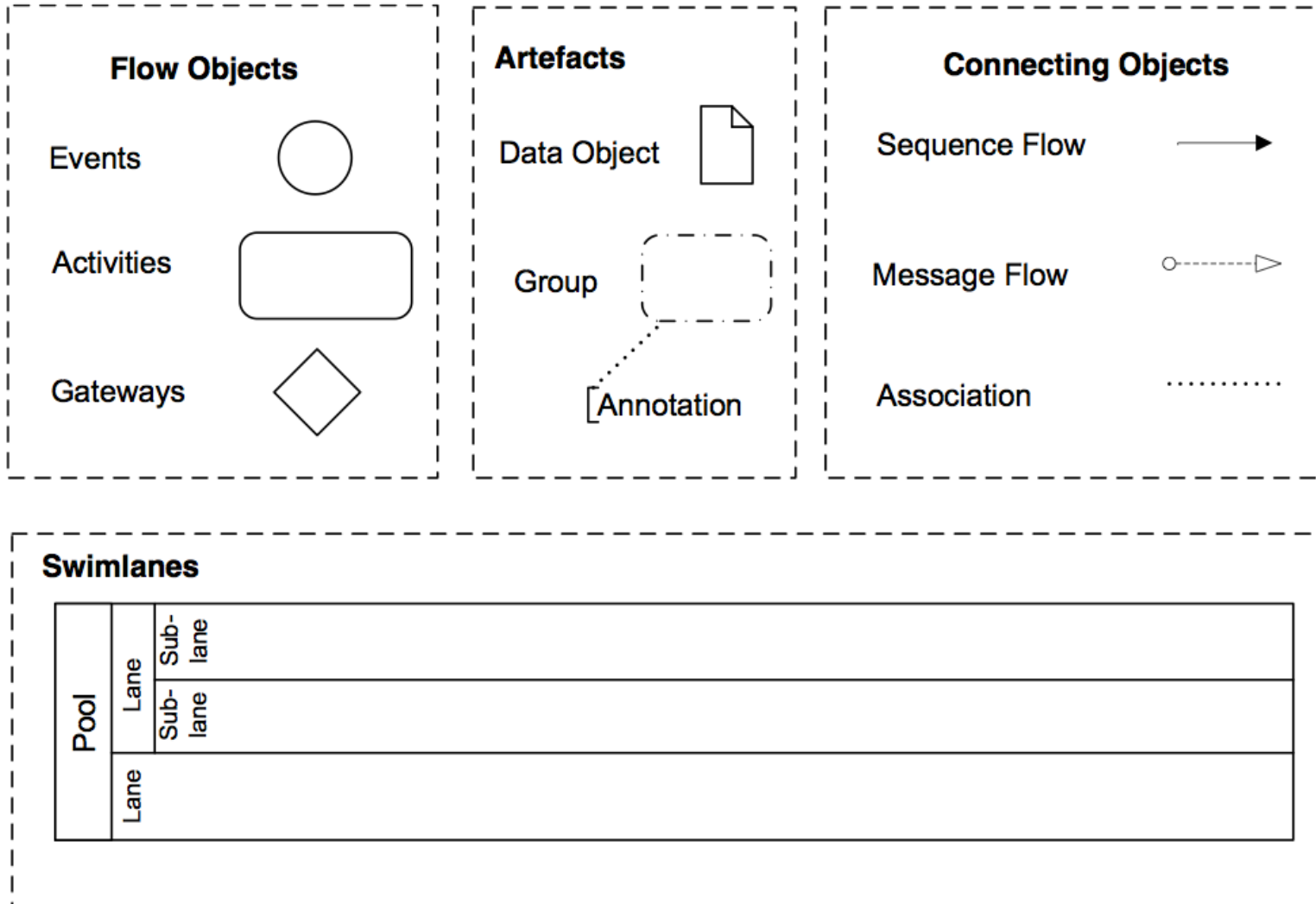
- Every month, a woman sits down and writes two words on 60 sheets of paper. Why does she do this?
- *Answer:* The woman owns her own business with 60 employees. Every month, she signs her name on their paychecks.



Business Process Model and Notation

- The primary goal of BPMN is to provide a notation that is **readily understandable by all business users**, from the **business analysts** that create the initial drafts of the processes, to the **technical developers** responsible for implementing the technology that will perform those processes, and finally, to the business people who will **manage and monitor** those processes. Thus, BPMN creates a **standardized bridge** for the gap between the business process design and process implementation.

Categories of elements



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Fig. 4.76. BPMN: categories of elements

Order process example

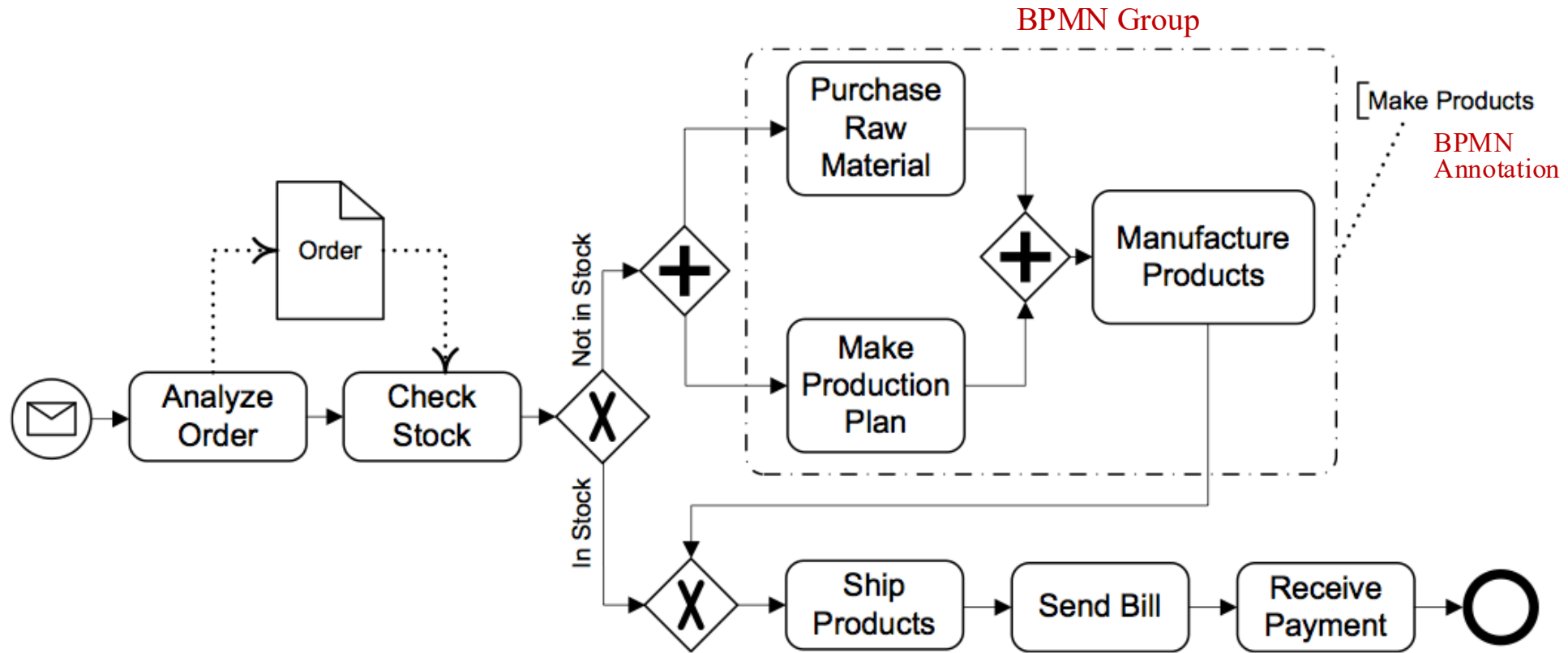
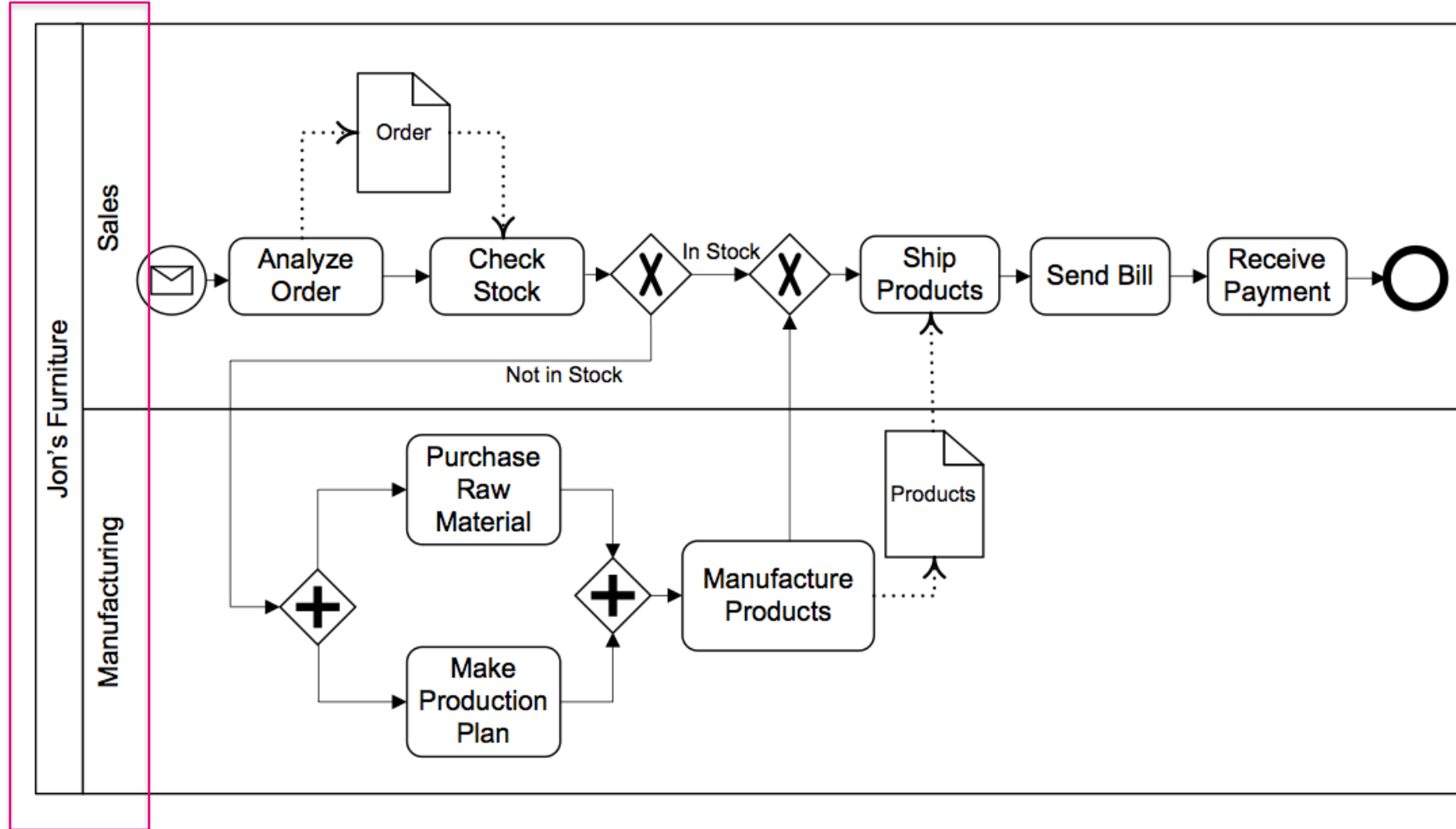


Fig. 4.77. Business process diagram expressed in BPMN

BPMN Text Annotation & BPMN Group

- **BPMN Group** is a visual container to use in creating informal, arbitrary grouping for elements in a business process diagram. You can create a Group to contain elements that belong together based on some common characteristic.
- A **BPMN Text Annotation** is simply a note – a note to process element or to the business process itself. You can use Text Annotation to include additional information to flow objects. Because Text Annotation is presented directly on the diagram, readers can easily see the information without the need to drill-down into the specification of elements.

BPMN roles



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Fig. 4.78. Business process diagram with role information

Activity types

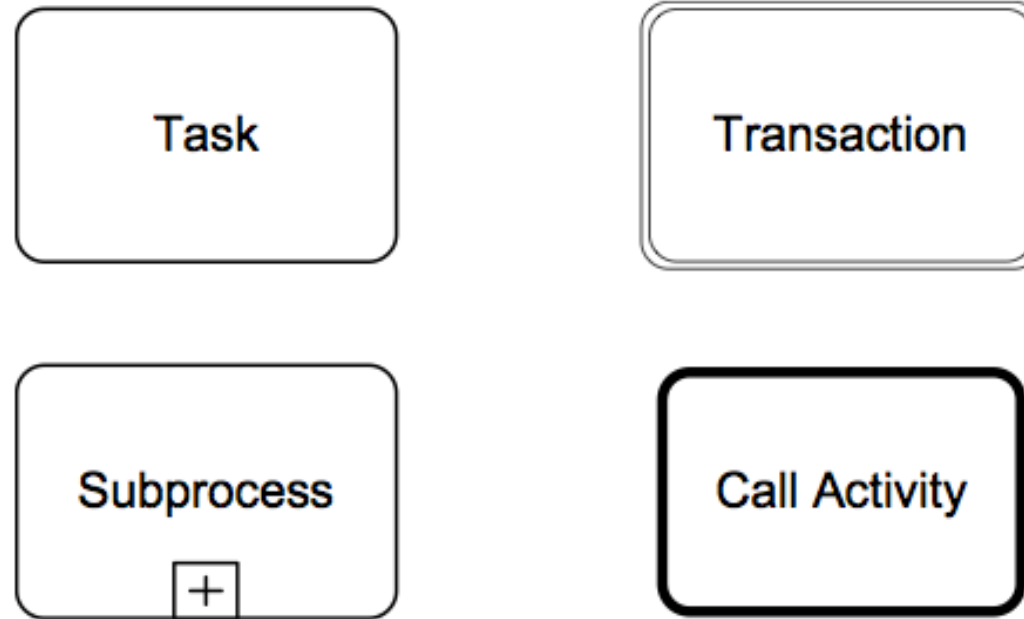


Fig. 4.79. Activity types in the BPMN

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Collapsed and expanded subprocess

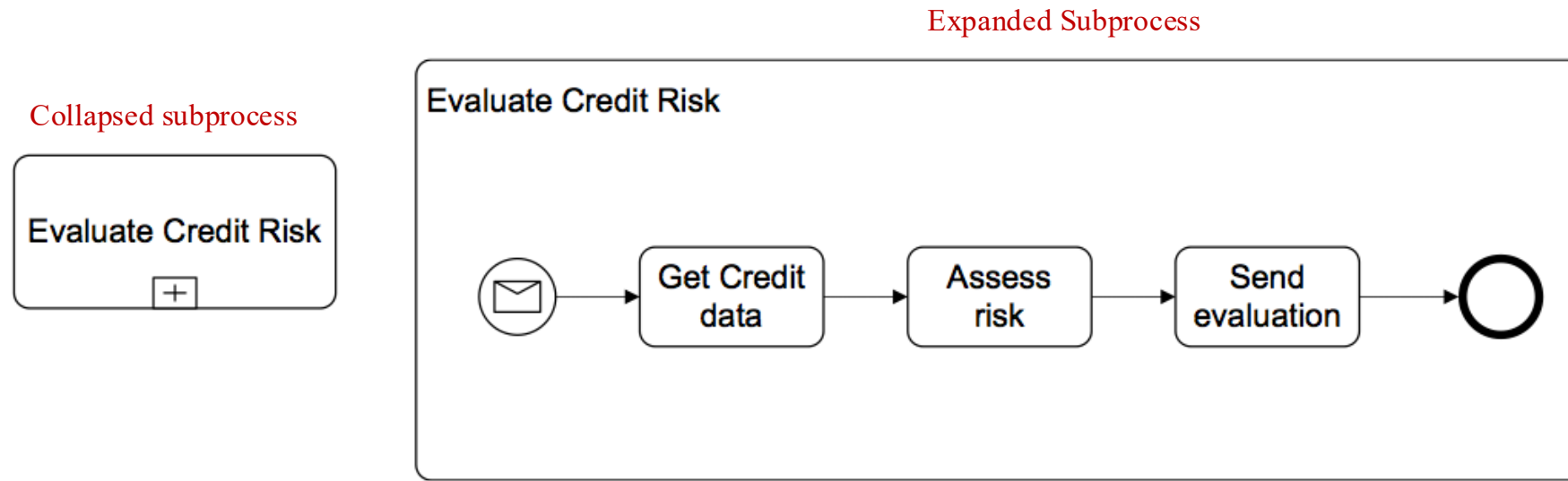
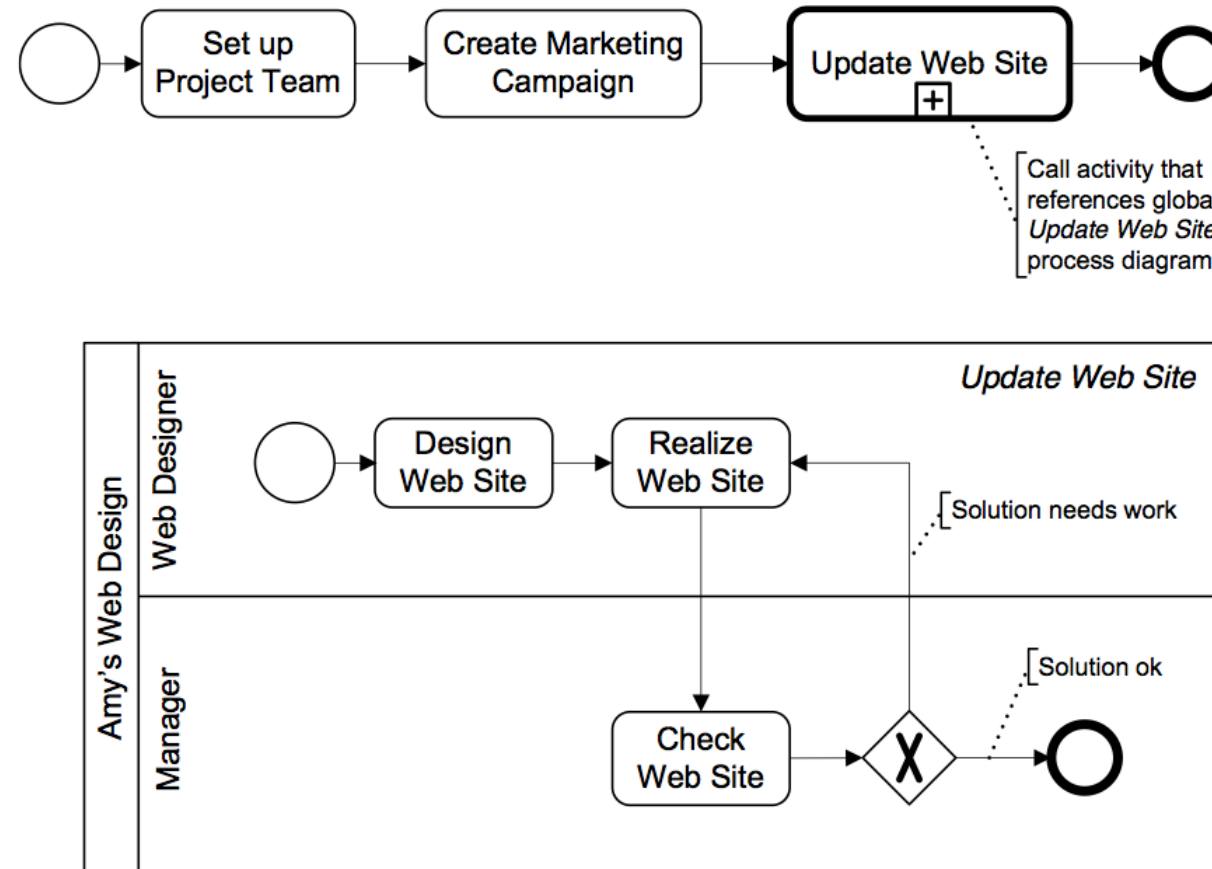


Fig. 4.80. Collapsed and expanded subprocess

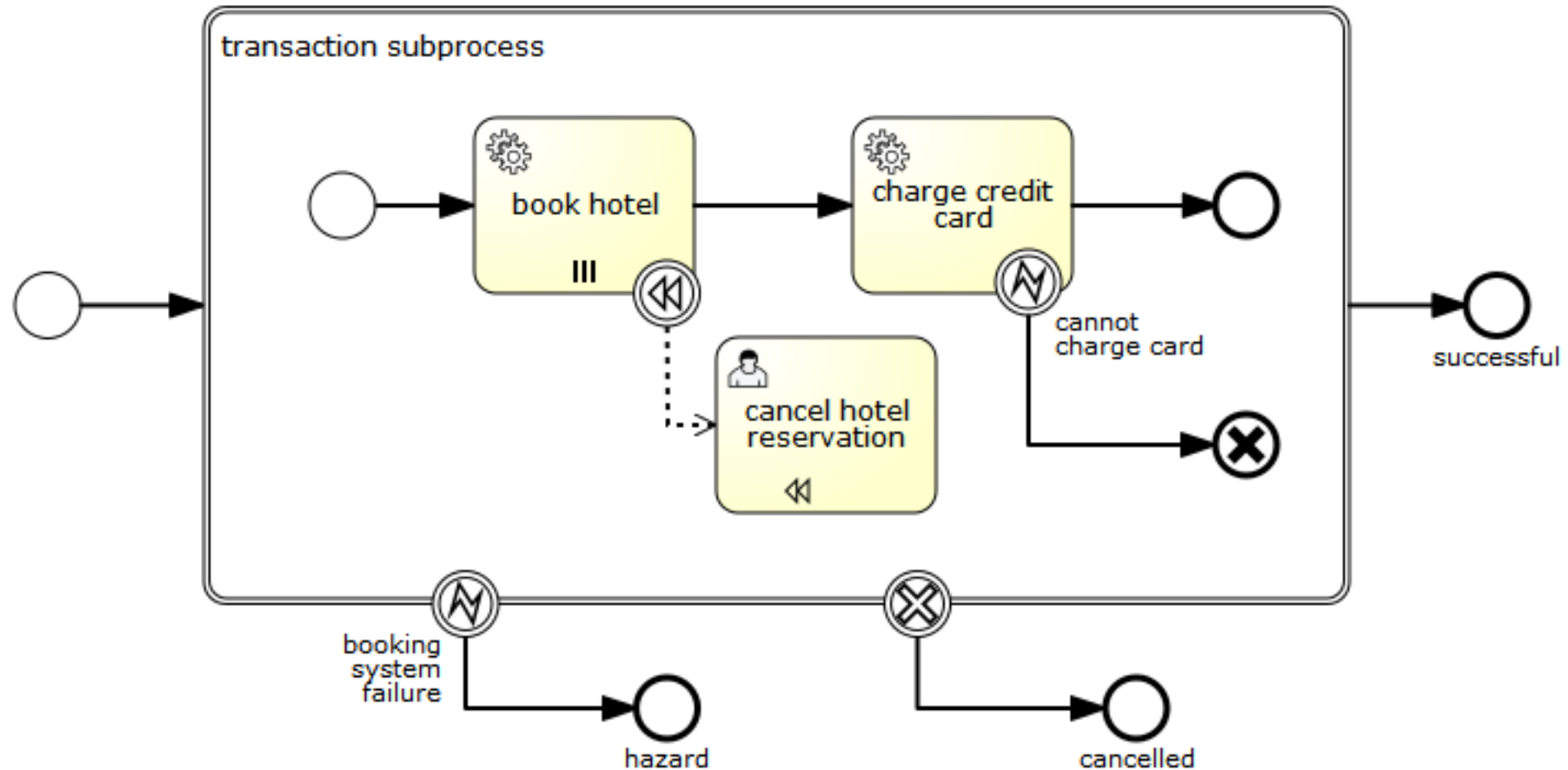
Call activity



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Fig. 4.81. Process diagram with a call activity that references a global process diagram; the reference is maintained in the respective attribute of the call activity

Transaction activity



BPMN activities

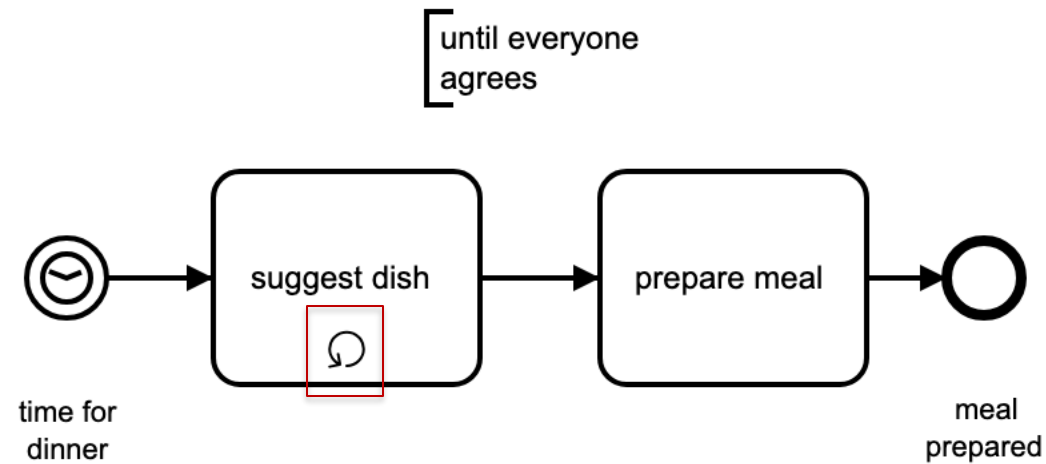
- Activities might also have an internal structure, in which case they are called **subprocesses**. Rather than showing the structure, the modeller can decide to hide the complexity of the subprocess, using the **plus symbol**. But subprocesses can also be expanded, exposing their internal structure.
- **Call activities** can be used to refer to **globally defined process diagrams**, or tasks, facilitating **reuse of activities**. The process is externalized (i.e. stored as separated BPMN) and can be invoked by different processes.
- In simple terms, a **Transaction activity** is a special type of activity that represents a group of actions or tasks that need **to be completed as a whole**. It's like a "mini-process" within a larger process that has to either **succeed completely or fail completely**.

Activity marker

-  Subprocess Marker
-  Loop Marker
-  Parallel MI Marker
-  Sequential MI Marker
-  Adhoc Marker
-  Compensation Marker

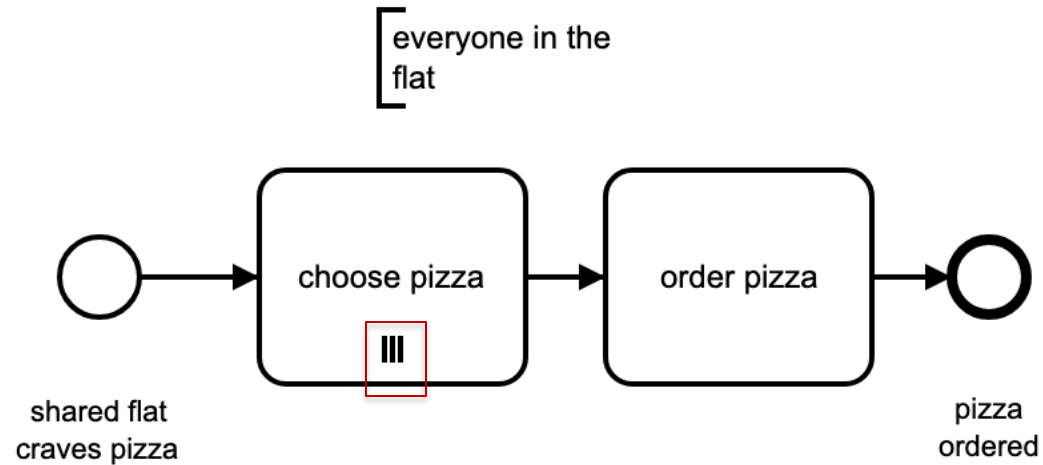
Fig. 4.82. Activity markers refine the behaviour of activities

Loop



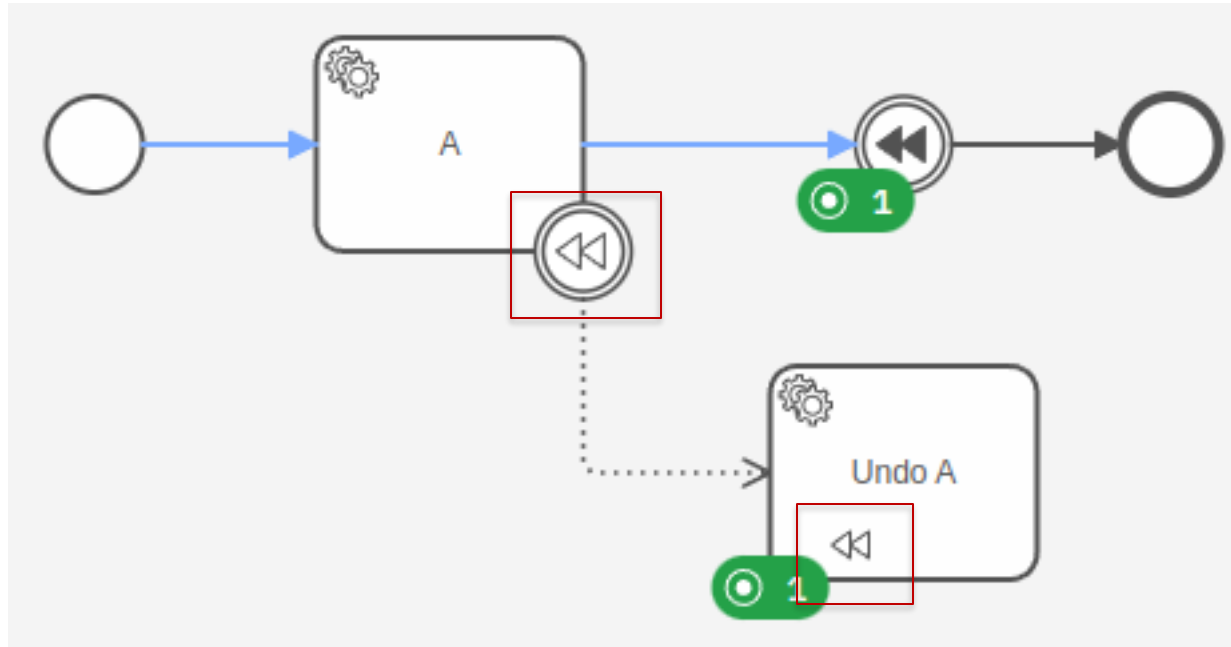
- A loop task repeats until a defined condition either applies or ceases to apply. Perhaps we suggest various dishes to our dinner guests until everyone agrees. Then, we can prepare the meal

Multiple instance (parallel)



- The individual cycles of a loop task must follow each other. If for example we live in a flat-sharing community and the roommates feel like eating pizza, the “choose pizza” task must be repeated for each roommate before we can order. A multiple task instantiates repeatedly and can be executed in sequence or in parallel

Compensation activity



- Compensation events assist with undoing steps that were already successfully completed in the case that their results are no longer desired and need to be reversed.

Adhoc process

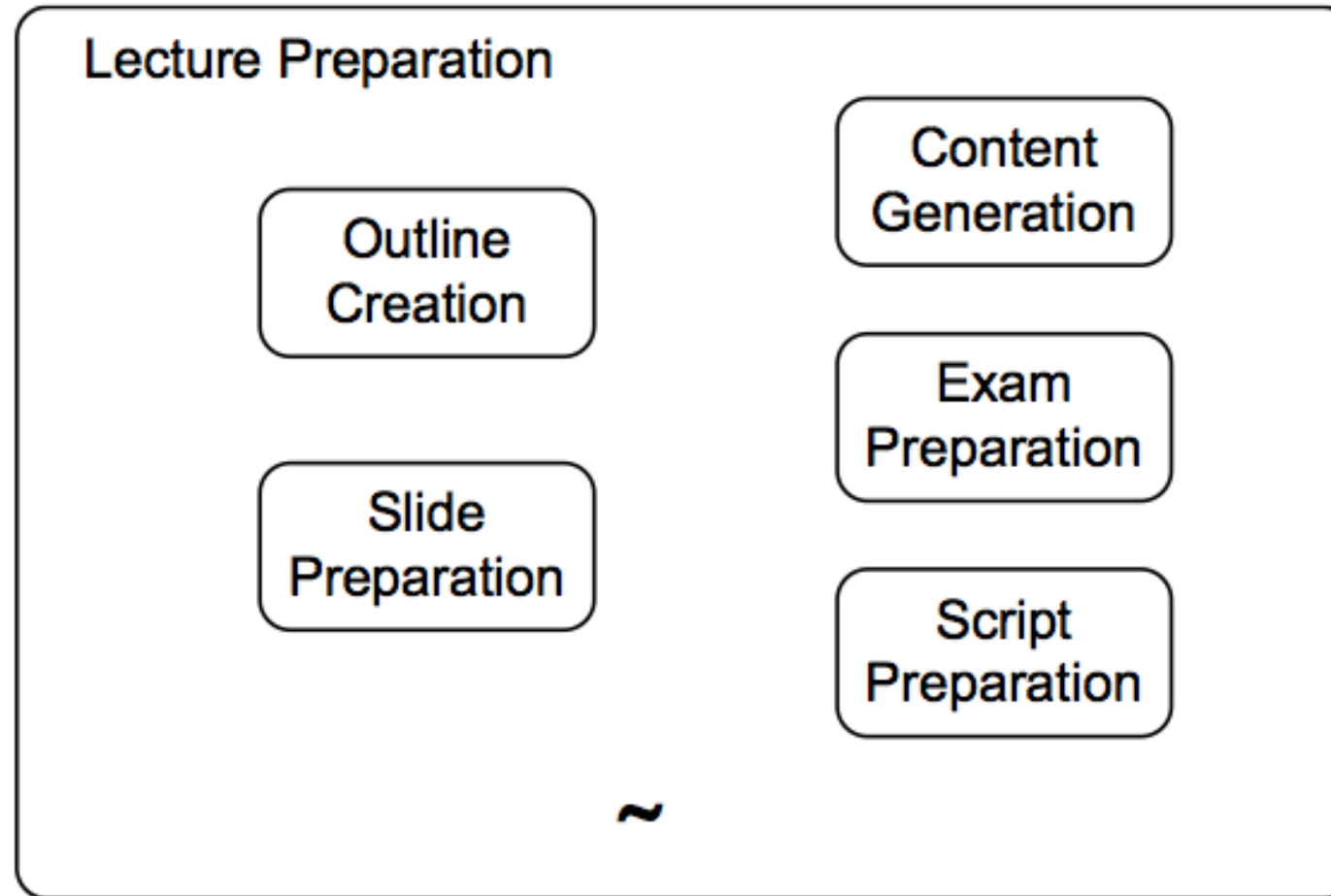


Fig. 4.83. Sample adhoc process

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The ad-hoc subprocess

- Use the ad-hoc subprocess to mark a segment in which the contained activities (tasks or subprocesses) can be:
 - Executed in any order,
 - Executed several times, or
 - Skipped

Task types

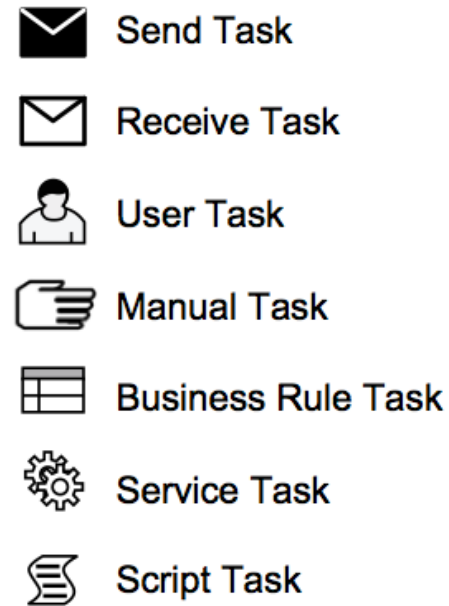
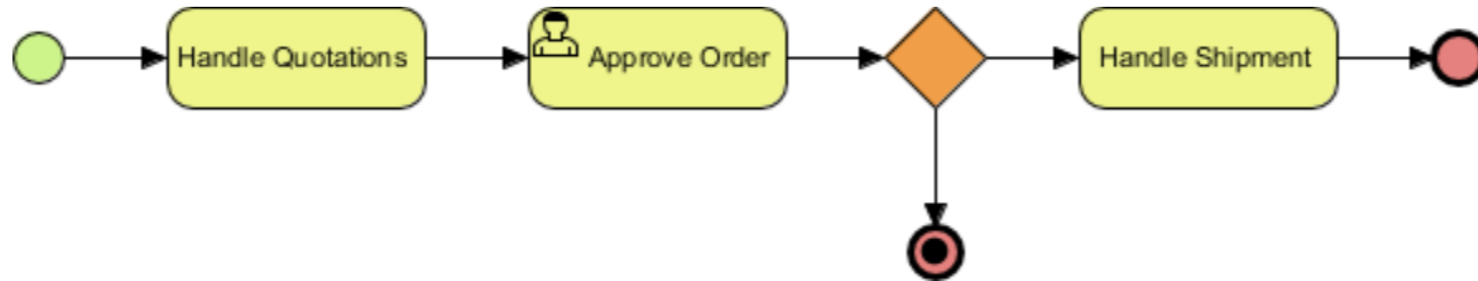


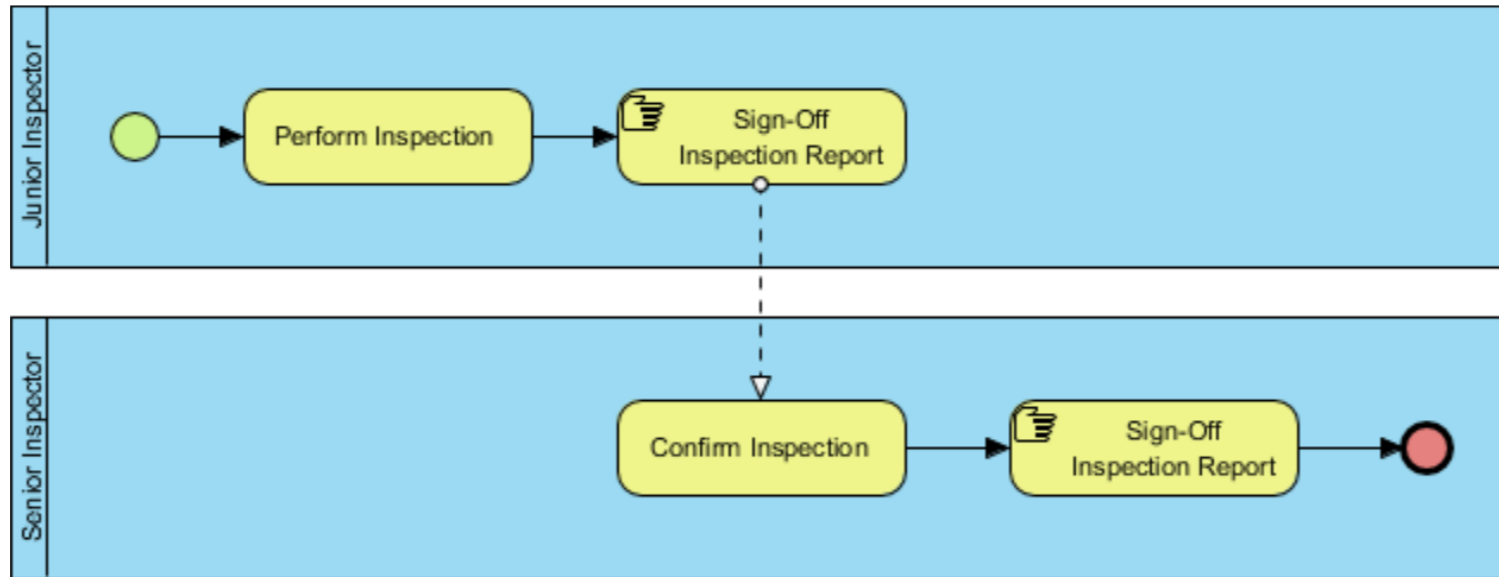
Fig. 4.84. Task types specify the kind of task that is represented

User task



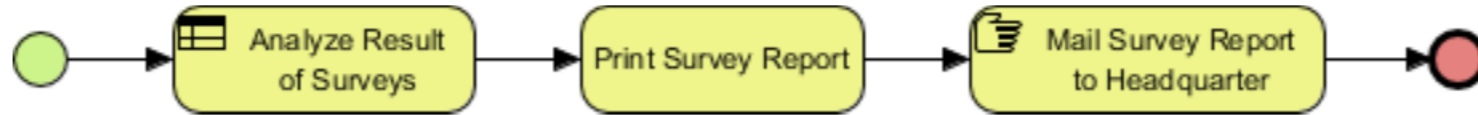
- A User Task represents that a human performer performs the Task with the use of a software application

Manual task



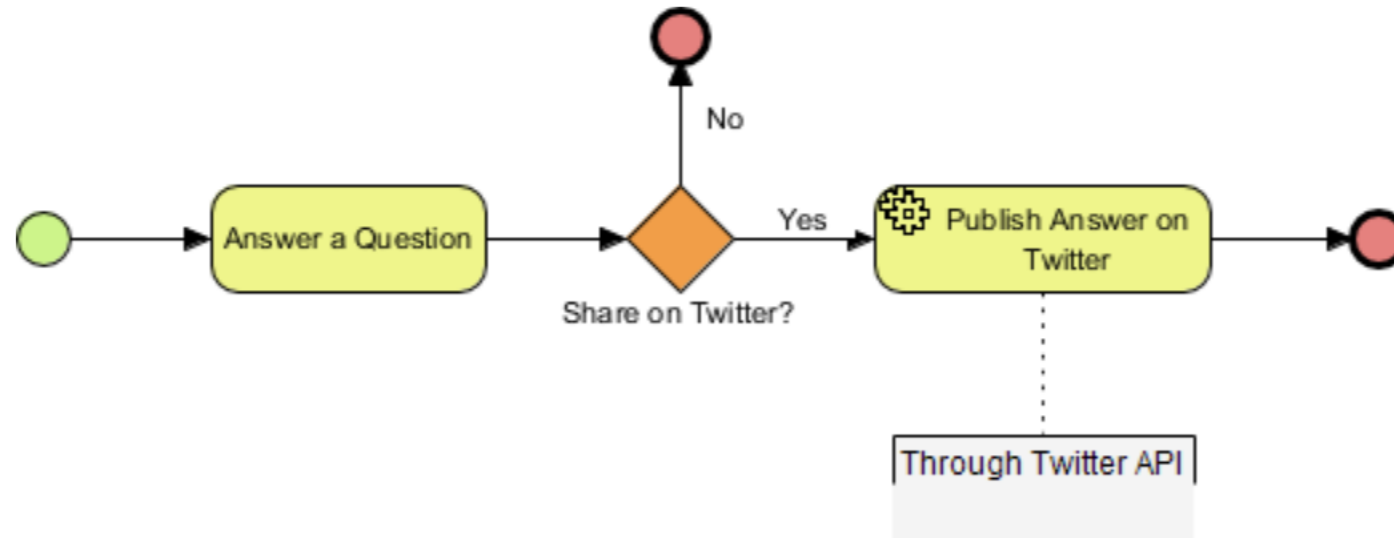
- A Manual Task is a Task that is performed without the aid of any business process execution engine or any application

Business rule task



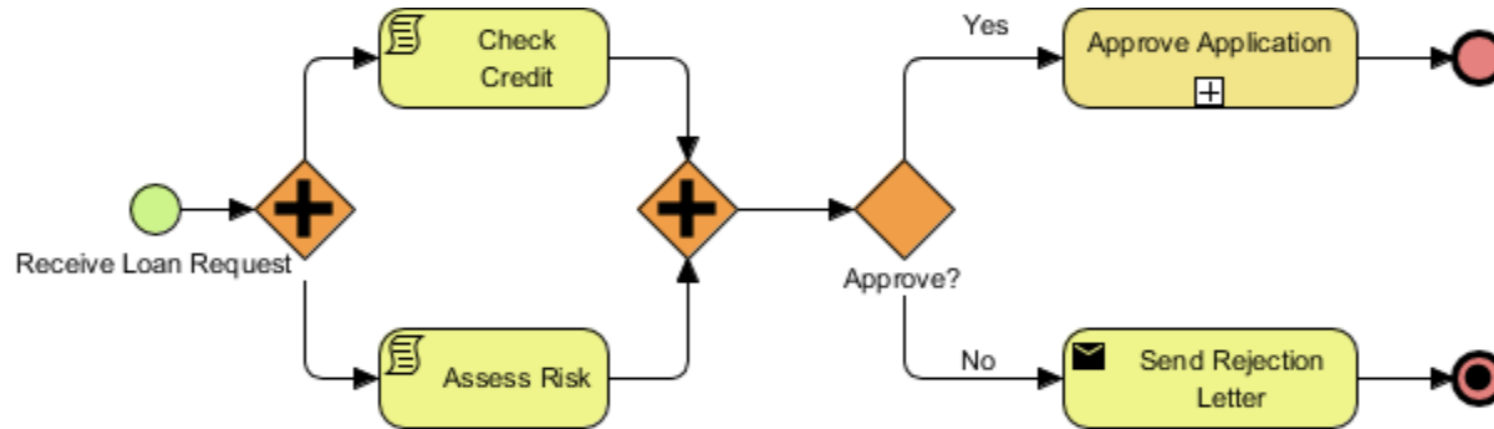
- The example below shows the use of Business Rule task in analyzing the result of surveys. It is expected that a business rule engine will be used in analyzing the data collected from the survey, and producing the result of analysis in return.
- The **business rule engine** is used to analyzing the data collected from the survey
- The advantage of using a **business rule task** is that the rules can be maintained and modified outside of the process model, making it flexible and easier to adapt to changing business requirements.

Service task



- A Service Task is a Task that uses a Web service, an automated application, or other kinds of service in completing the task

Script task



- A Script Task is executed by a business process engine. The task defines a script that the engine can interpret. When the task begins, the engine will execute the script. The Task will be completed when the script is completed

Events

	Start Events	Intermediate Events				End Events
	Catching	Catching	Boundary Interrupting, Catching	Boundary Non-Interrupting, Catching	Throwing	Throwing
None or blanco: Untyped events, indicate start point, state changes or final states.						
Message: Receiving and sending messages.						
Timer: Cyclic timer events, points in time, time spans or timeouts.						
Escalation: Escalating to a higher level of responsibility.						
Conditional: Reacting to changed business conditions or integrating business rules.						
Link: Off-page connectors. Two corresponding link events equal a sequence flow.						
Error: Catching or throwing named errors.						
Cancel: Reacting to cancelled transactions or triggering cancellation.						
Compensation: Handling or triggering compensation.						
Signal: Signalling across different processes. A signal thrown can be caught multiple times.						
Multiple: Catching one out of a set of events. Throwing all events defined.						
Parallel Multiple: Catching all out of a set of parallel events.						
Terminate: Triggering the immediate termination of a process.						

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Fig. 4.85. Common event types in the BPMN, adapted from the BPMN Poster, BPM Offensive Berlin (2011)

Types of events

- Events that occur at the beginning of the process are called **Start events**(thin narrow line) – and every process starts with an event.
- Events that occur in the middle of the process (between activities) are called **Intermediate events** (*thin double line*).
- Events that occur at the end of the process are called **End events**(thick single line) and every process has at least one end event.
- All of these can be catching and throwing events:
 - **Catching events** are events with a defined trigger. They are started once the trigger has activated or fired
 - **Throwing events** are triggers for catching events and are triggered by the process

Throwing and catching events

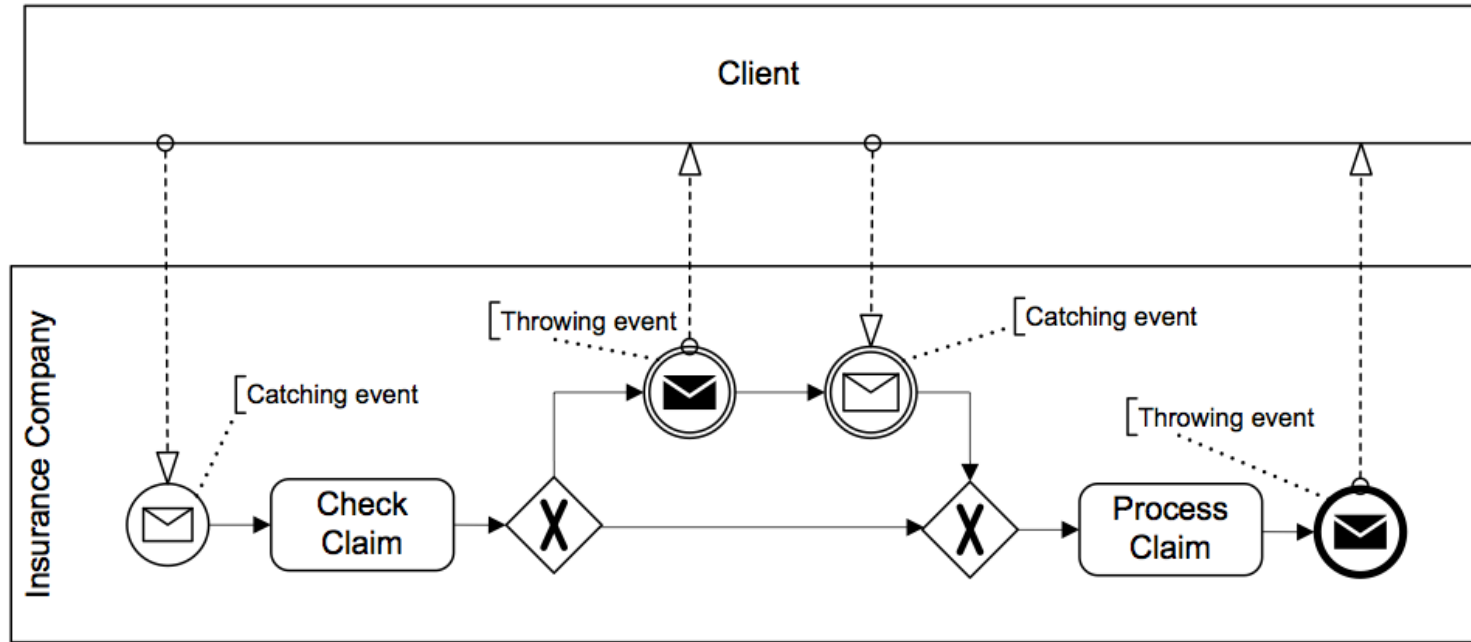


Fig. 4.86. Throwing and catching events

Using markers to identify send tasks and receive tasks

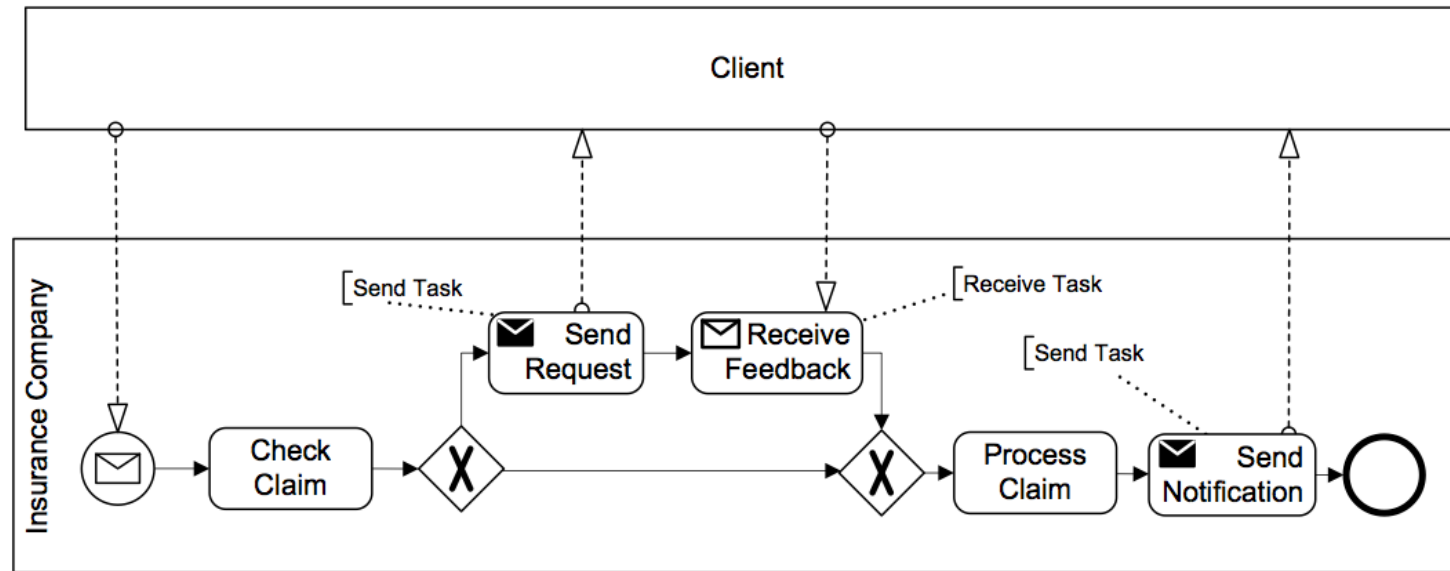
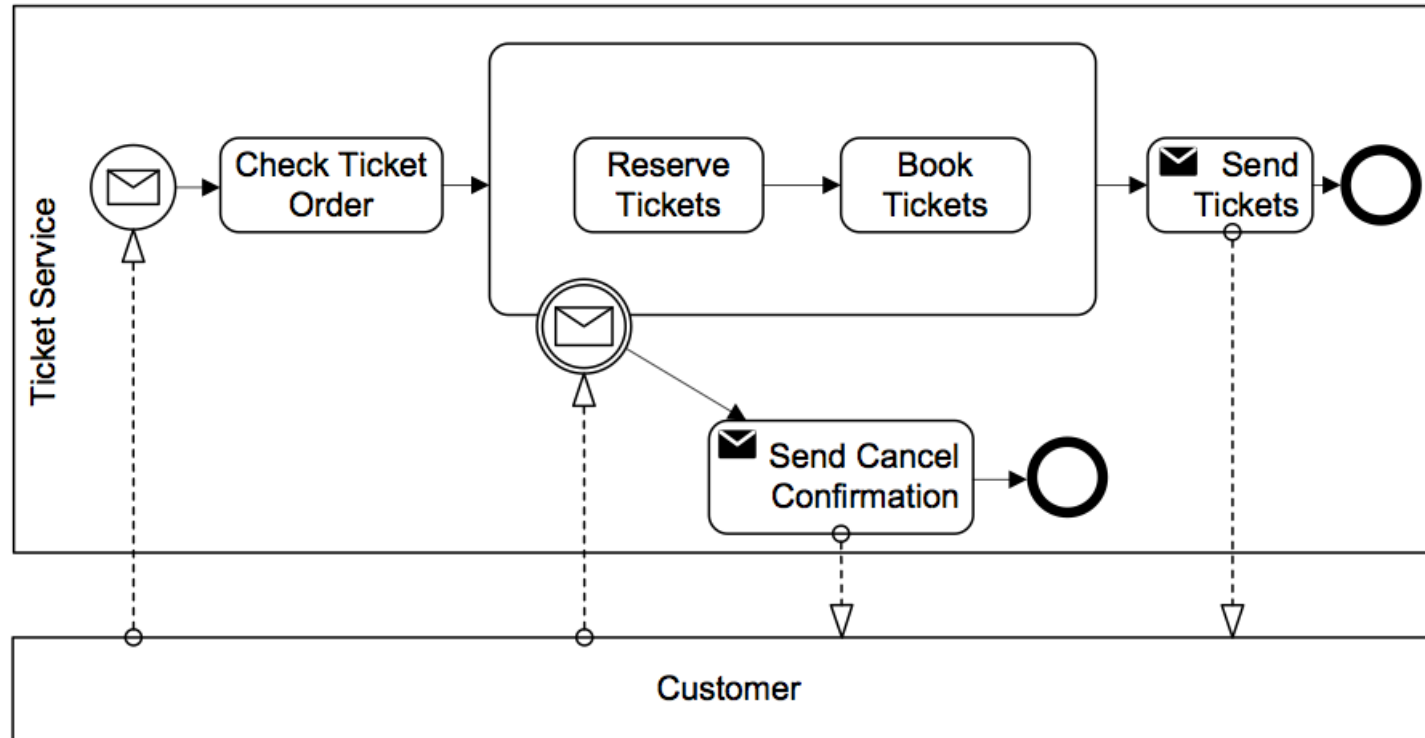


Fig. 4.87. Using markers to identify send tasks and receive tasks

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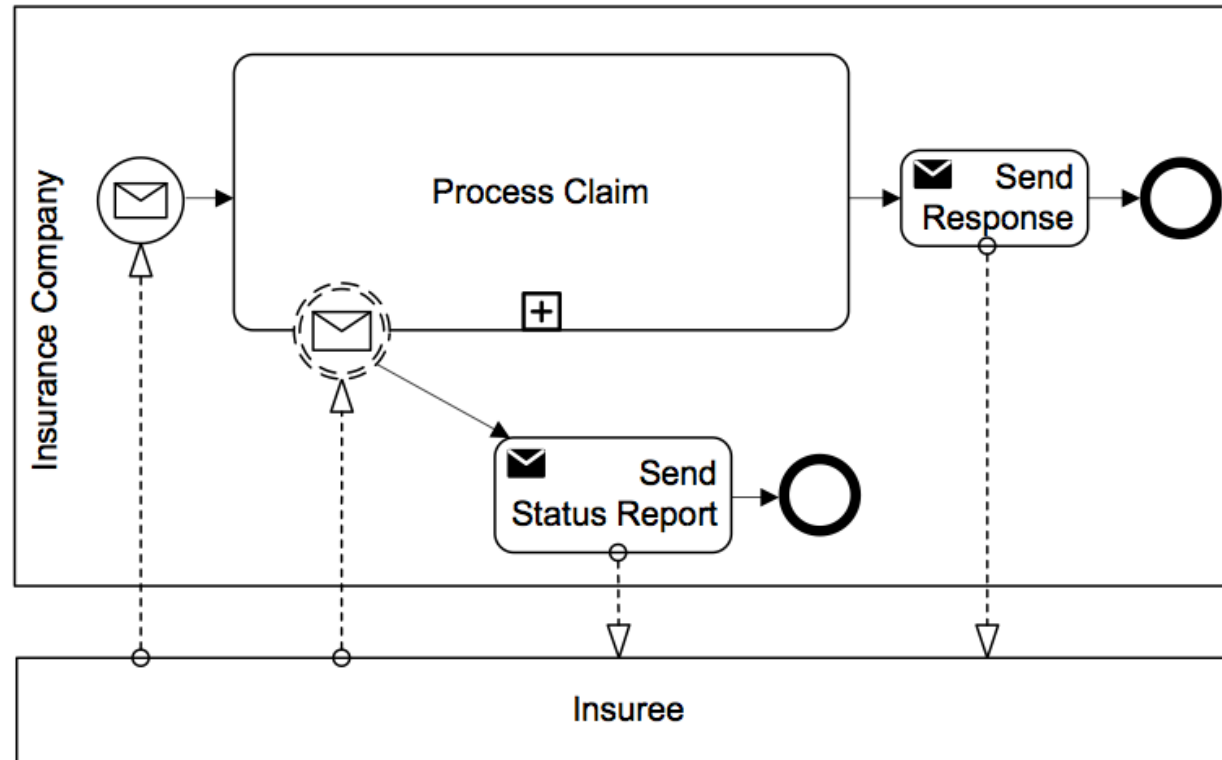
Process diagram with interrupting boundary event



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Fig. 4.88. Process diagram with interrupting boundary event

Process diagram with non-interrupting boundary event



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Fig. 4.89. Process diagram with non-interrupting boundary event

Process diagram with interrupting and non-interrupting boundary timer events

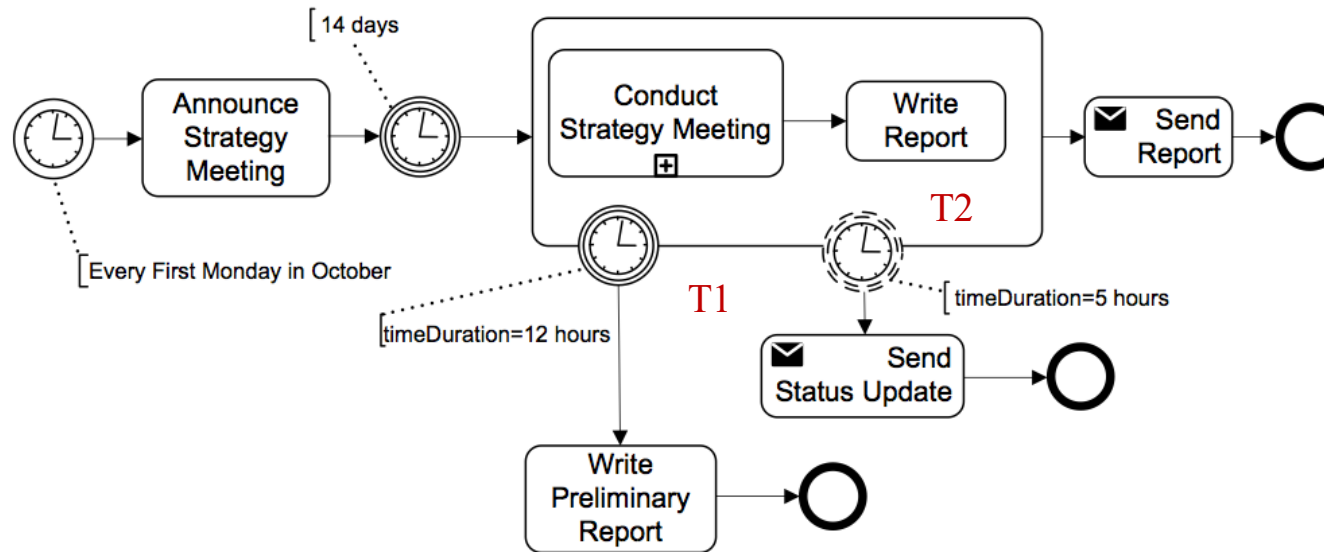


Fig. 4.90. Process diagram with interrupting and non-interrupting boundary timer events

- Assuming the subprocess is still active after 5 hours, Timer 2 is triggered, and a status update is sent (can also happen multiple times)
- If the subprocess is not completed after 12 hours, Timer 1 elapses, and the subprocess is interrupted. A preliminary report is written, and the process terminates

Link events connect different parts of one process

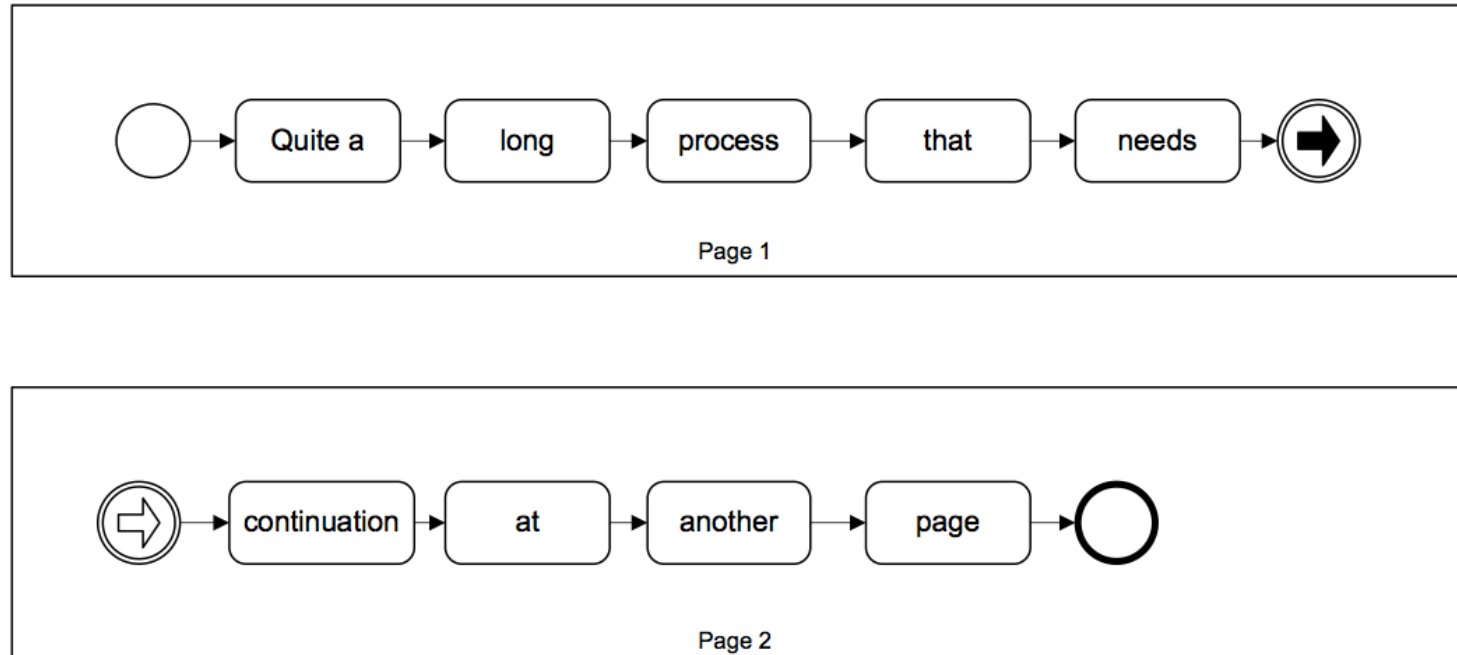


Fig. 4.91. Link events connect different parts of one process

Error thrown in a subprocess

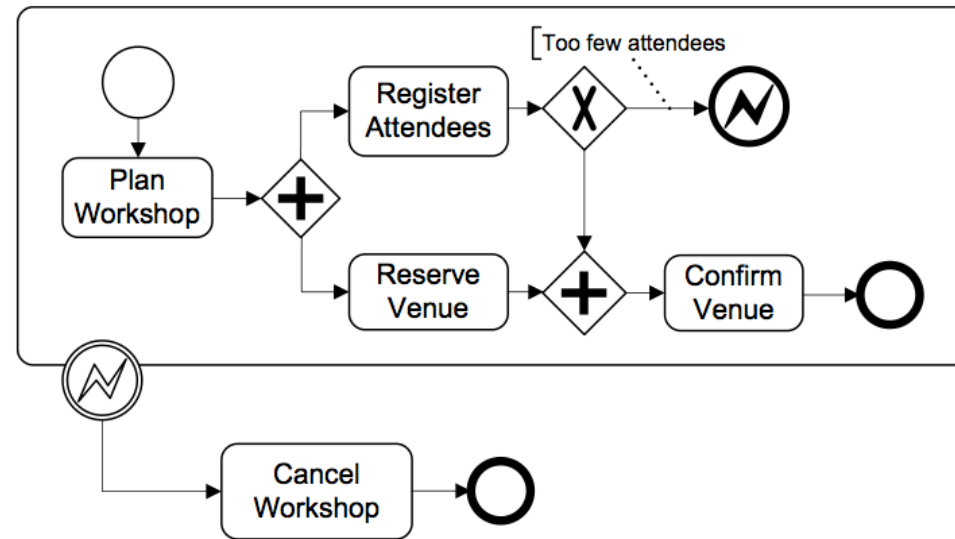
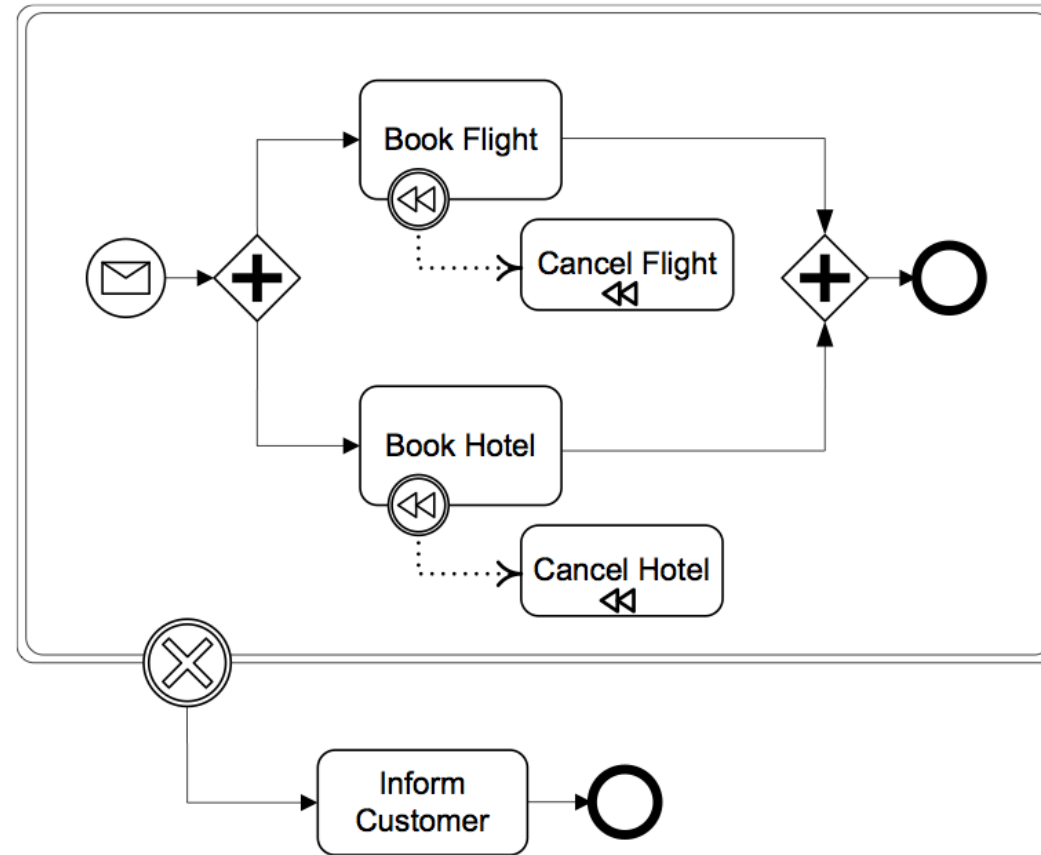


Fig. 4.92. An error is thrown in a subprocess; it is caught by an error boundary event attached to that subprocess

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- Notice that once the error event is thrown, any running activity in the subprocess will be interrupted.

Transaction and compensation elements



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Fig. 4.93. Business process diagram with transaction and compensation elements, adapted from Object Management Group (2011)

Gateways

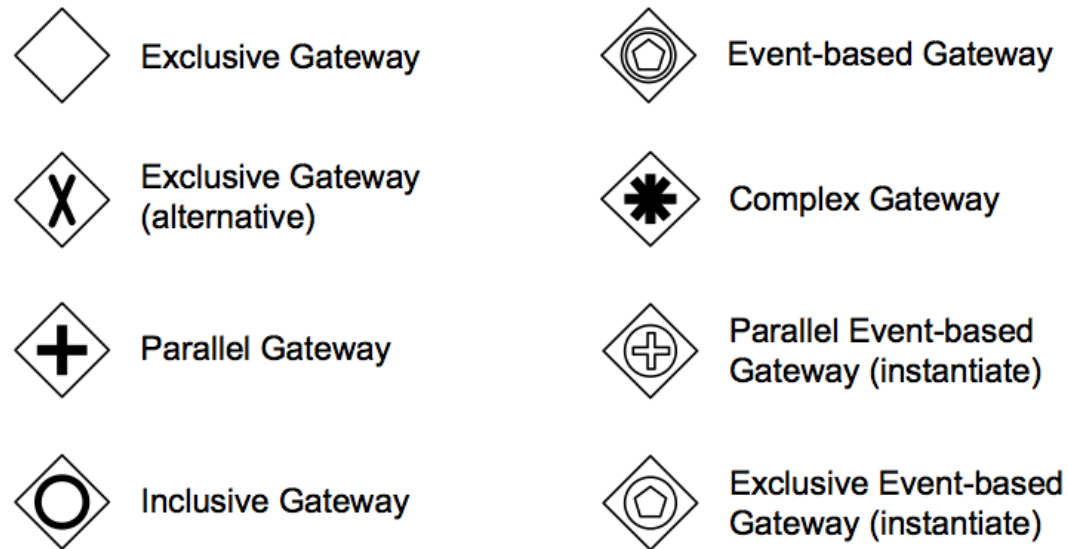


Fig. 4.94. Gateway types in the BPMN, Object Management Group (2011)

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- In BPMN, each gateway acts as a join node or as a split node

Parallel gateway

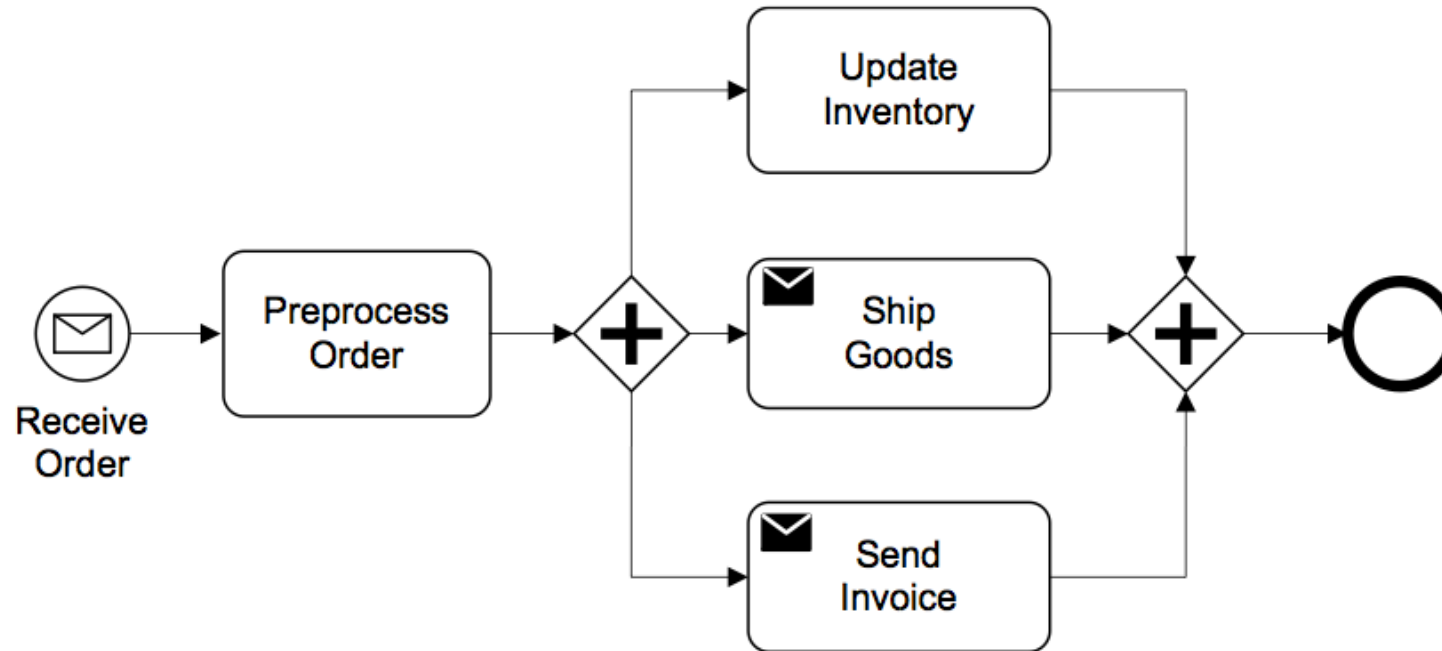


Fig. 4.95. Example involving the parallel gateway

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- The parallel gateway triggers the execution of three activities

Exclusive gateway

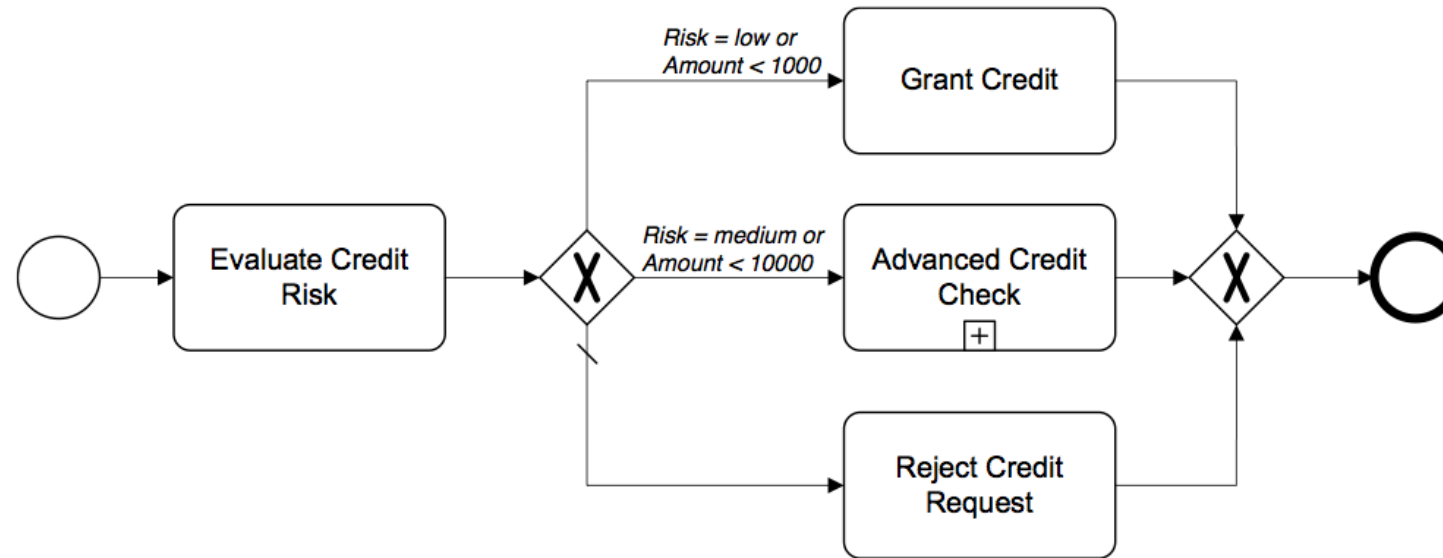


Fig. 4.96. Exclusive gateway with conditions and default flow

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- Conditions are “evaluated in order”
- Default (/) condition is always evaluated last

Loop (1/2)

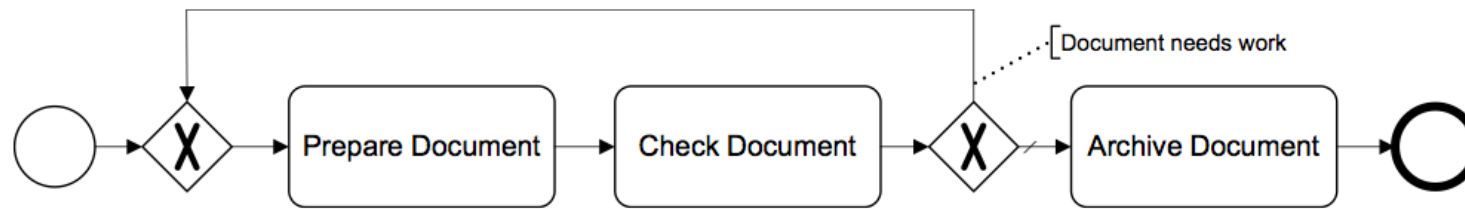


Fig. 4.97. Exclusive gateways realizing a loop

Loop (2/2)

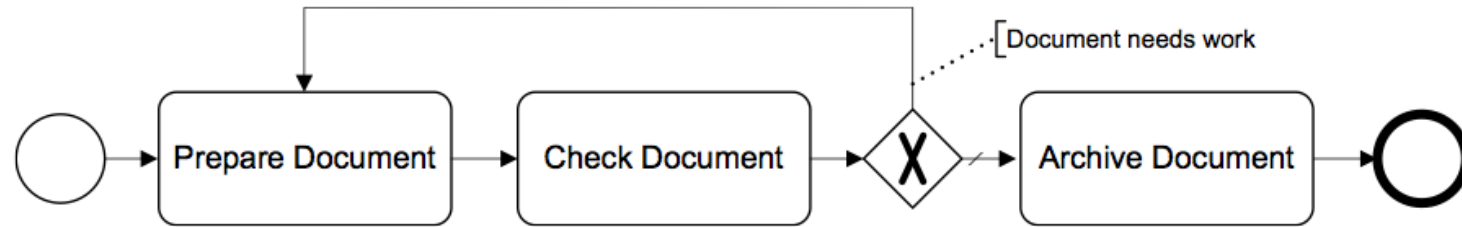


Fig. 4.98. Process diagram with uncontrolled flow

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- According to the BPMN, activities with multiple incoming edges act as merge nodes.

Livelock

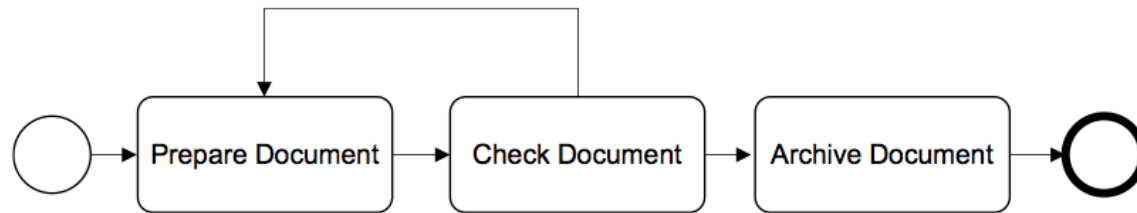


Fig. 4.99. Process diagram with split and join activities, representing a livelock

- Process activities with multiple outgoing edges are also possible in BPMN
- Activities with multiple outgoing edges represent a parallel split gateway
- These activities might lead to modelling errors

Event based gateway

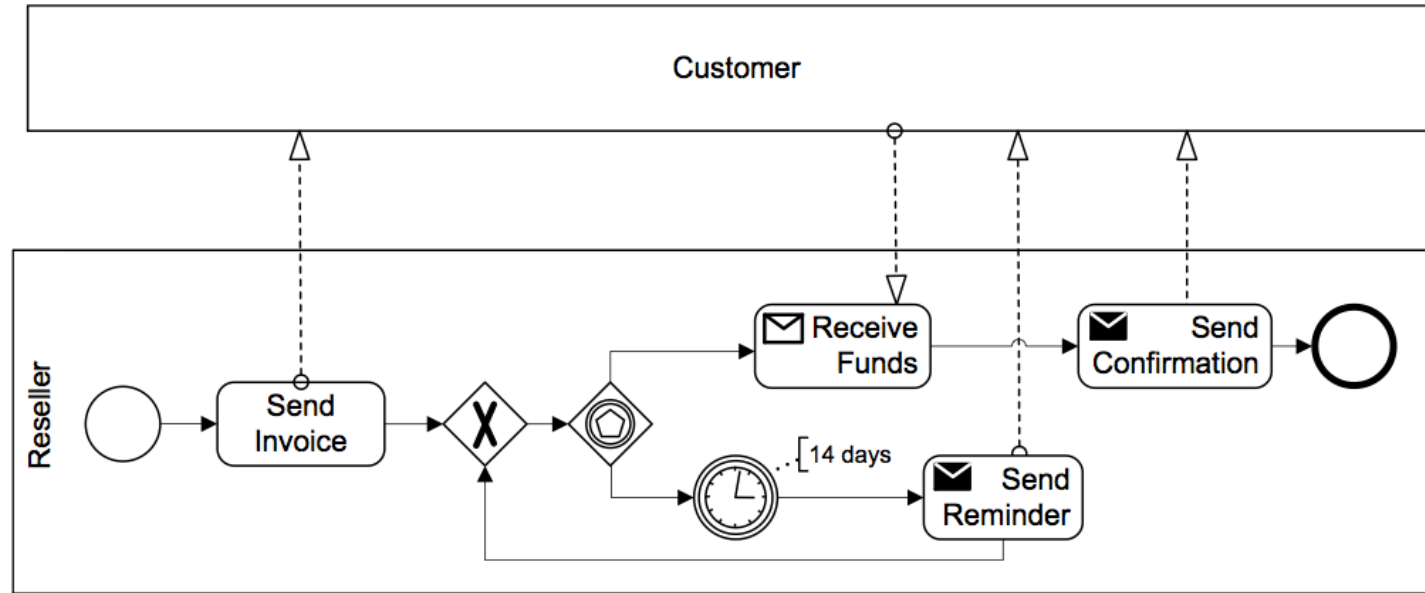


Fig. 4.100. Example of an *event-based* gateway

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- Used when the process must wait for **an external event** to determine the next step
- The path is chosen based on which **event occurs first**

Inclusive or gateway

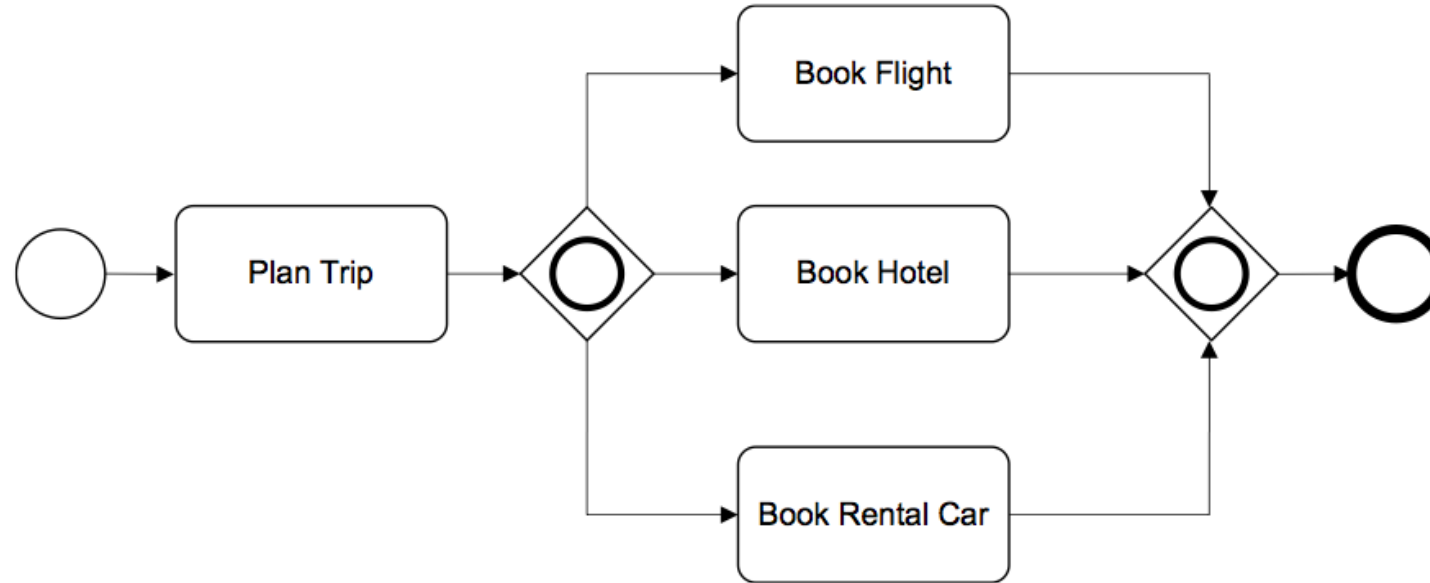


Fig. 4.101. Example of an *inclusive or* gateway

Handling Data

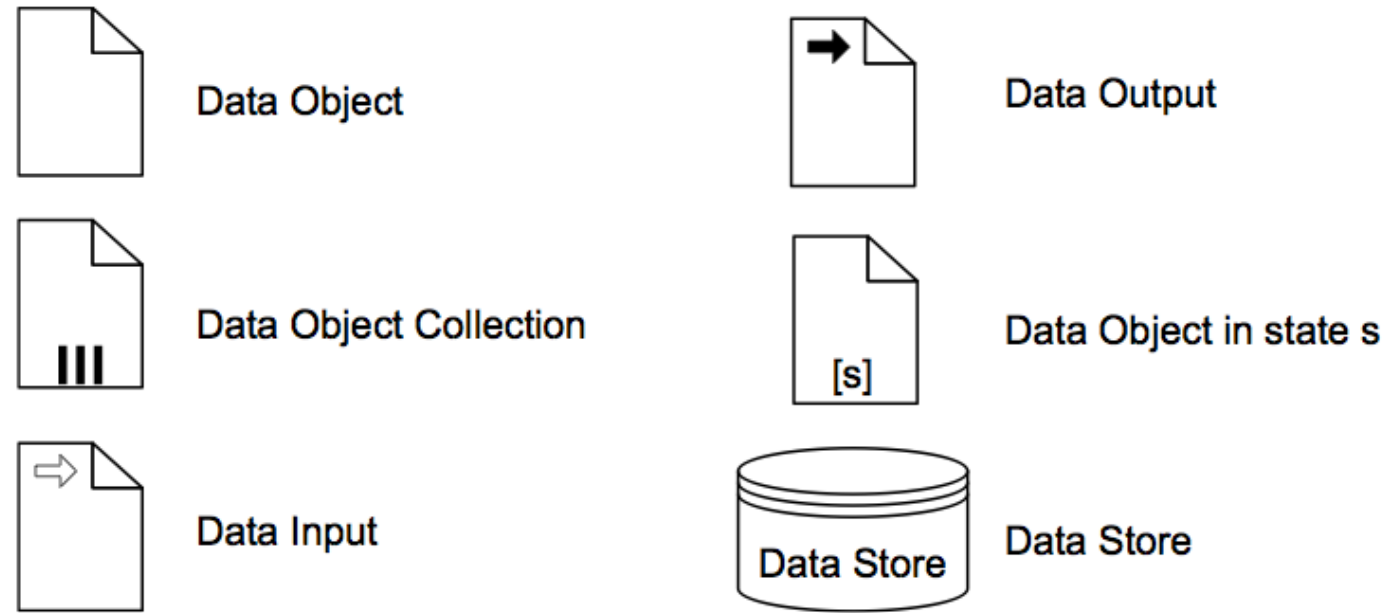
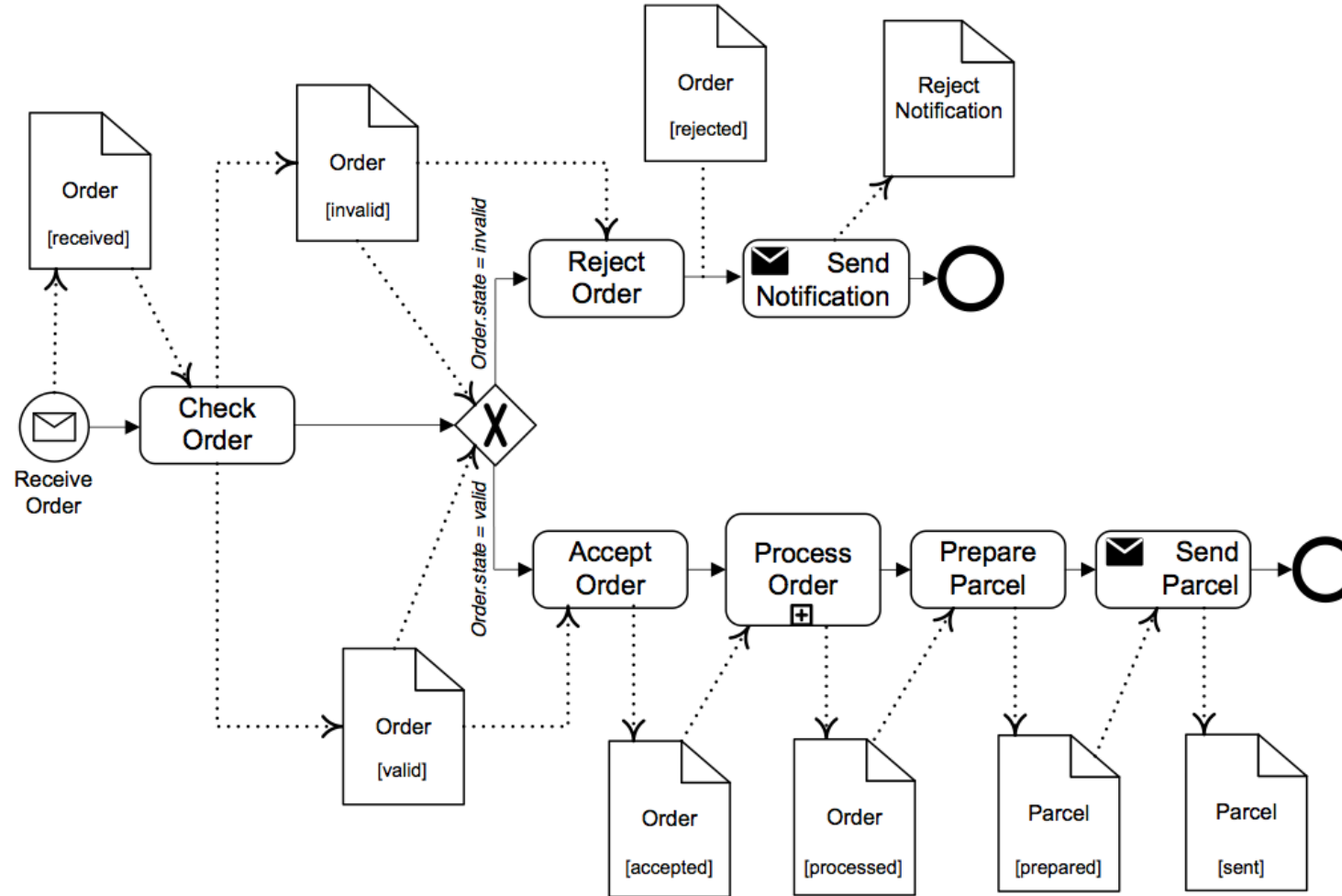


Fig. 4.102. Notational elements regarding data

Process diagram involving data objects and states



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Fig. 4.103. Process diagram involving data objects

Data object collections

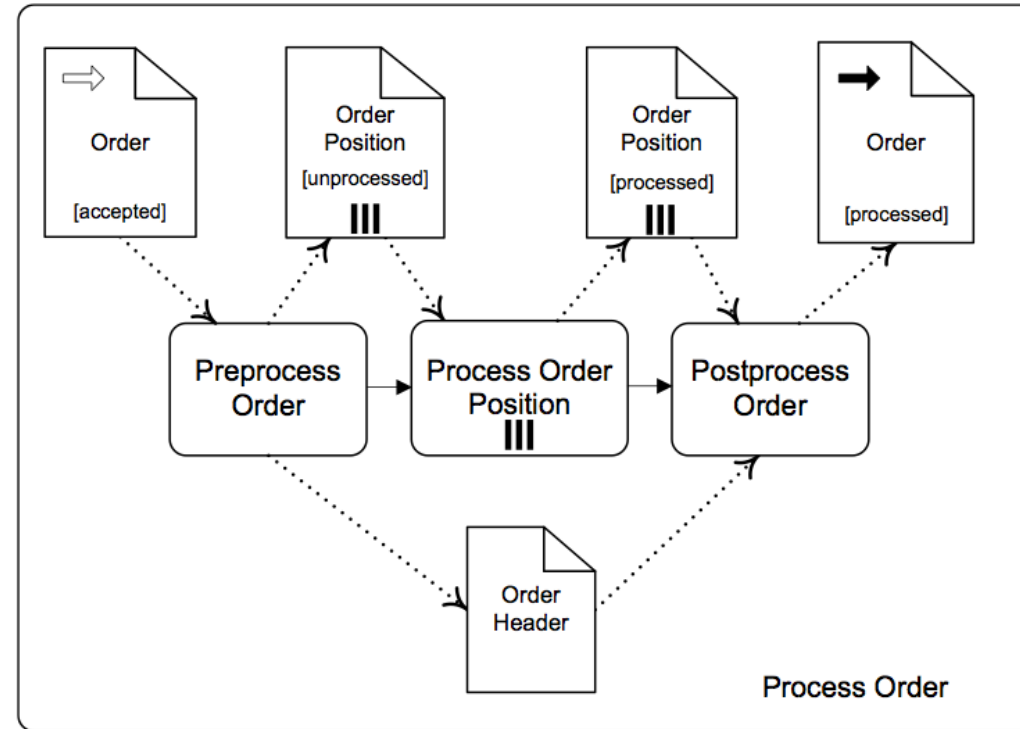
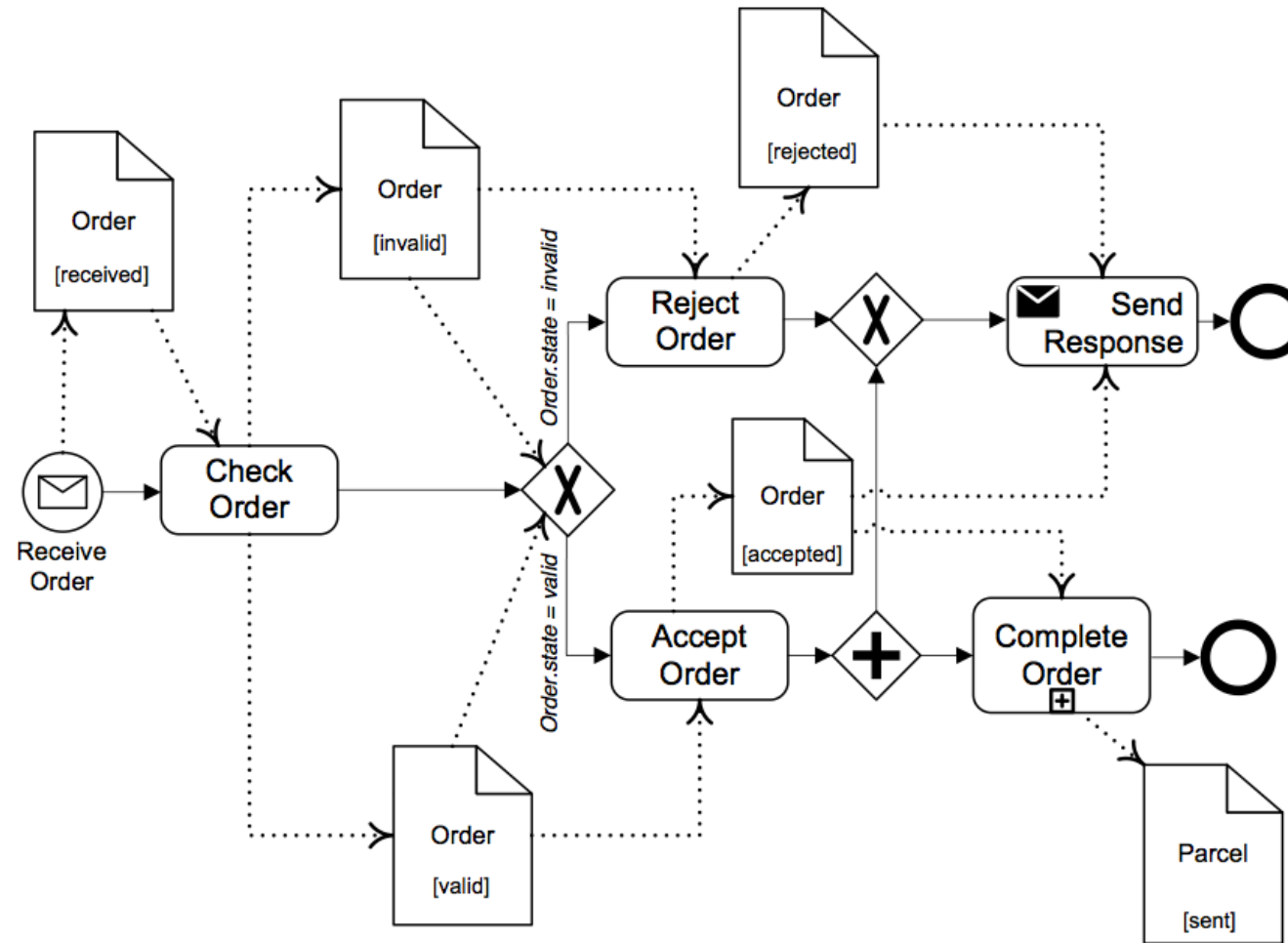


Fig. 4.104. Diagram of the *Process Order* subprocess from Figure 4.103, involving data object collections

Process diagram involving multiple input sets of an activity



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Fig. 4.105. Process diagram involving multiple input sets of an activity

Multiple start events

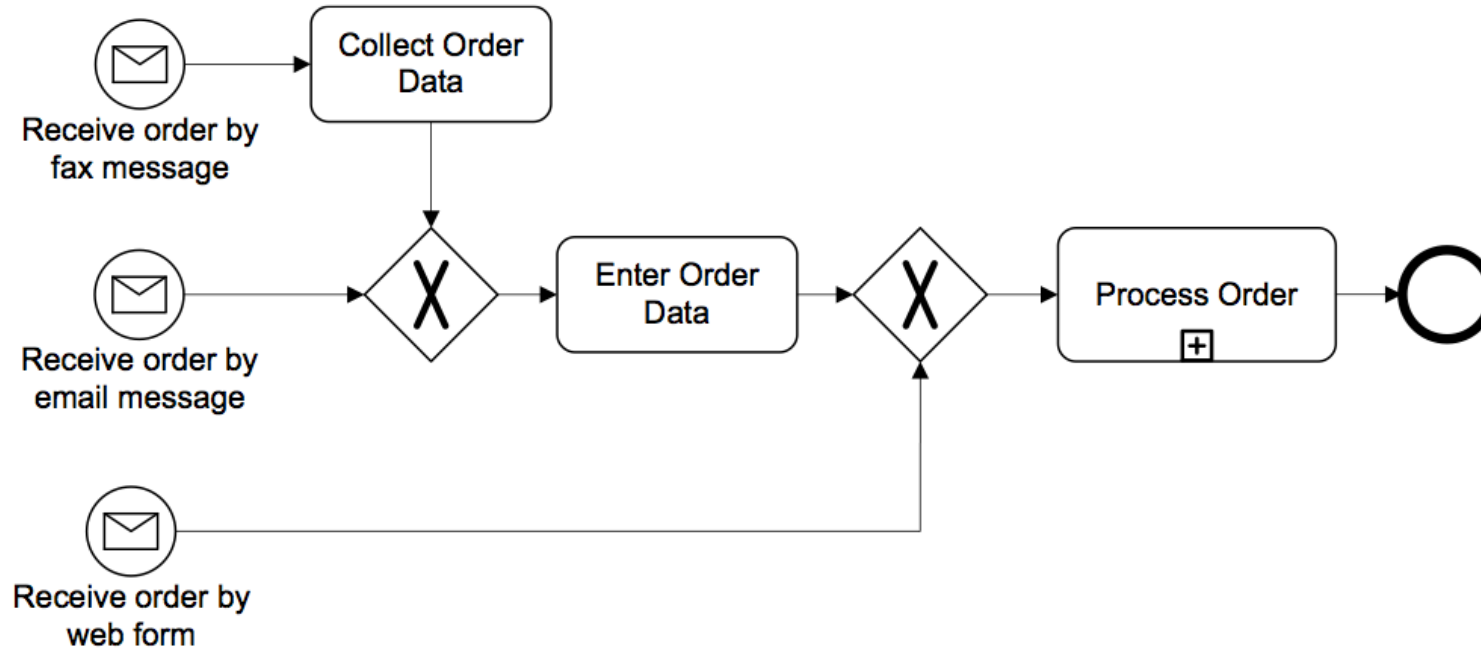


Fig. 4.106. Process diagram with multiple alternative start events

Process diagram with two start events, both of which need to occur to instantiate the process

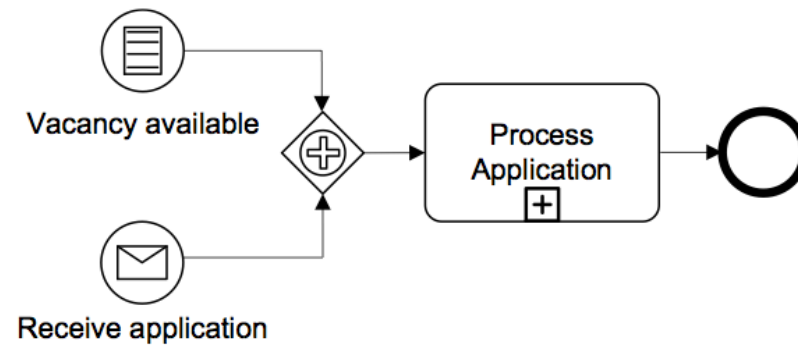


Fig. 4.107. Process diagram with two start events, both of which need to occur to instantiate the process

Collaborating Prozesse

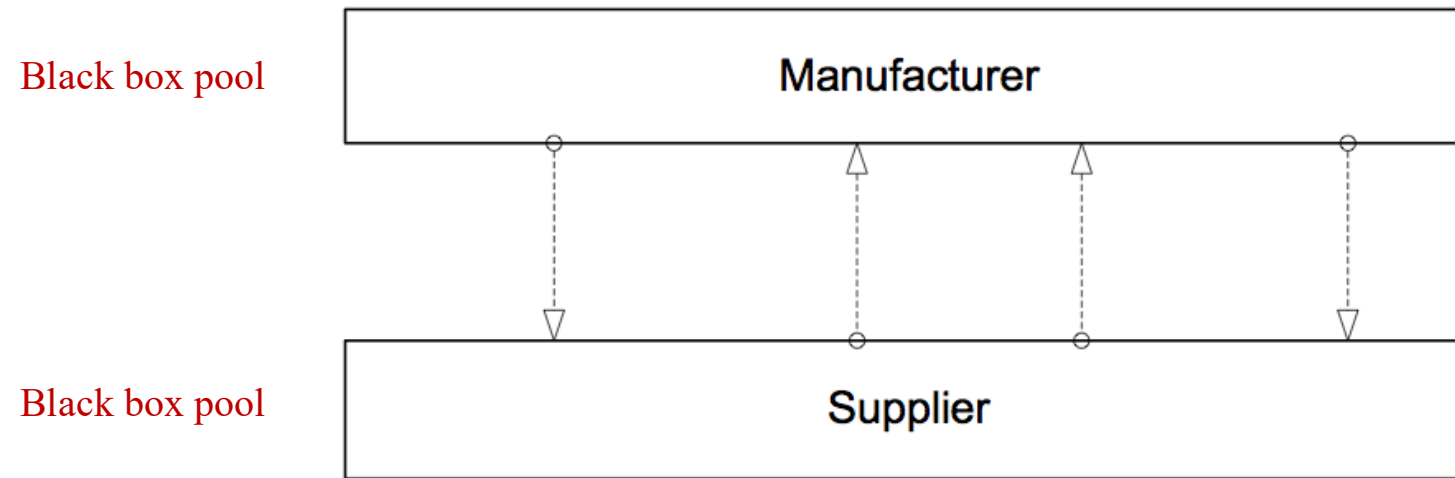


Fig. 4.108. Business processes collaborating through message flow

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Collaborating business processes with public process of the Supplier

Black box pool

White box pool

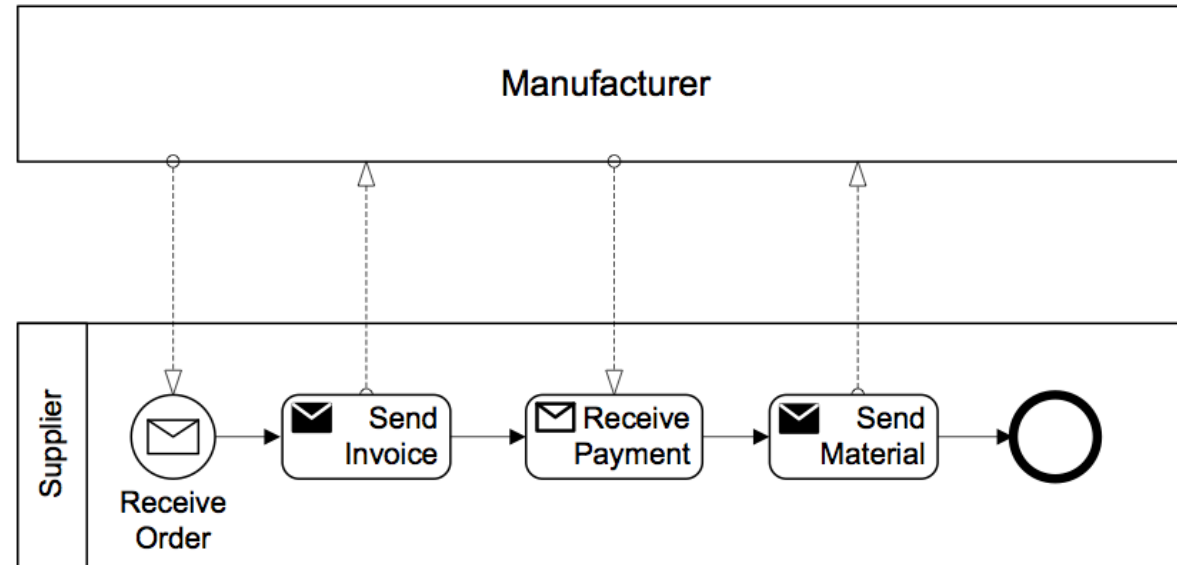


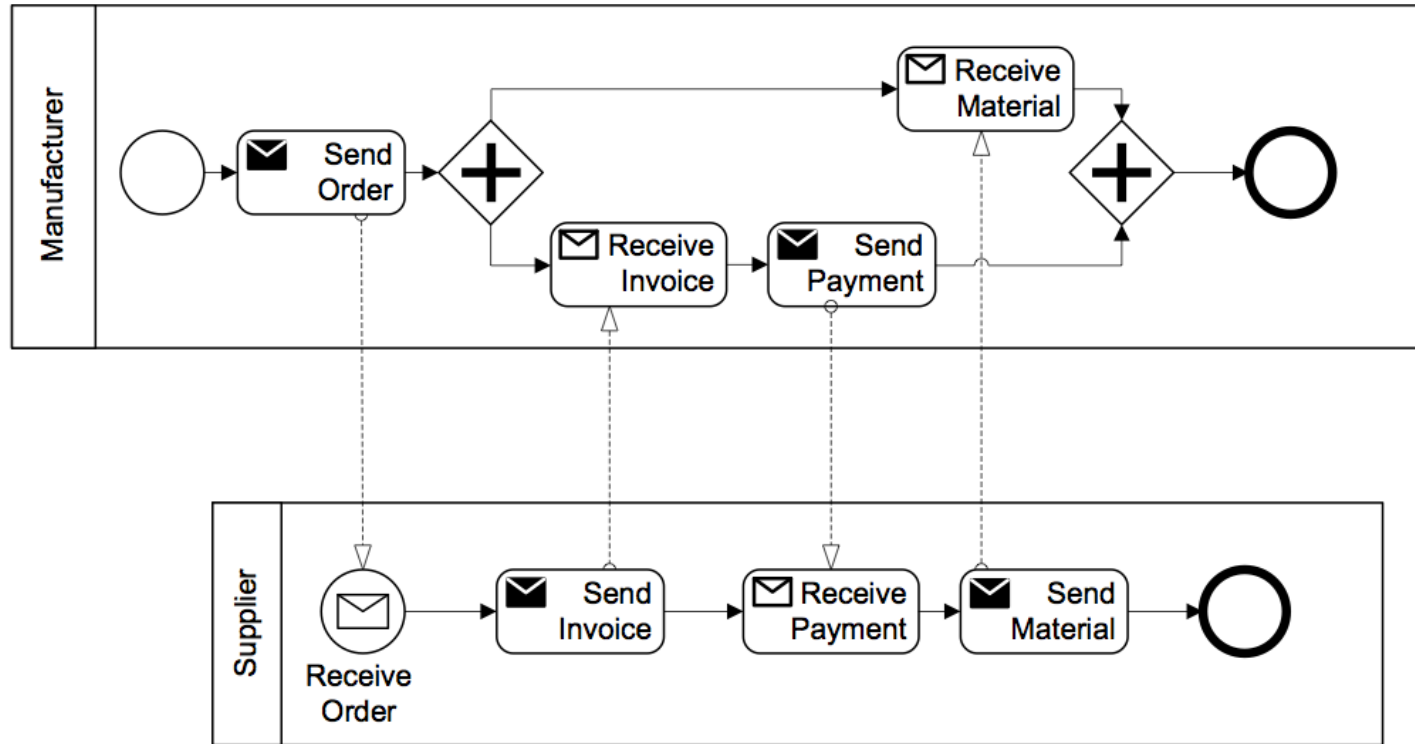
Fig. 4.109. Collaborating business processes with public process of the *Supplier*

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Collaborating business processes with public processes of both partners

Public process

Public process



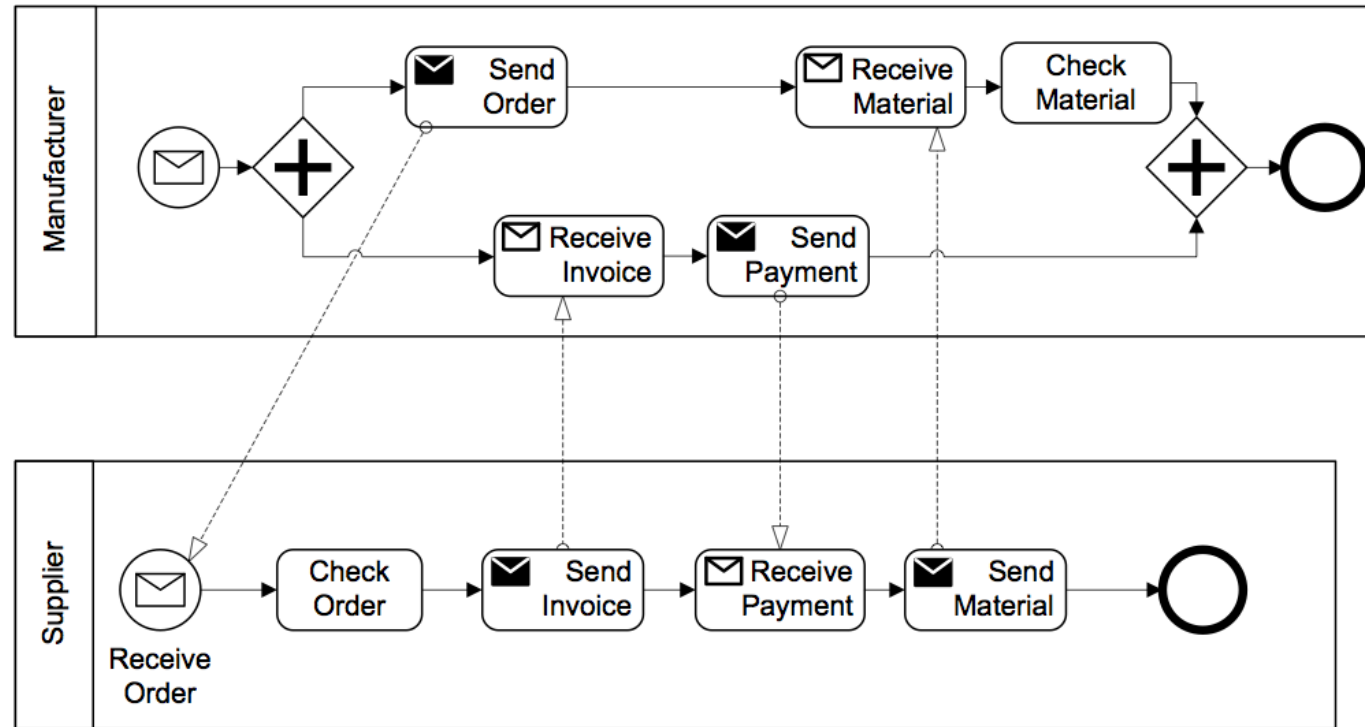
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Fig. 4.110. Collaborating business processes with public processes of both partners

Collaborating business processes with private processes of both partners

Private process

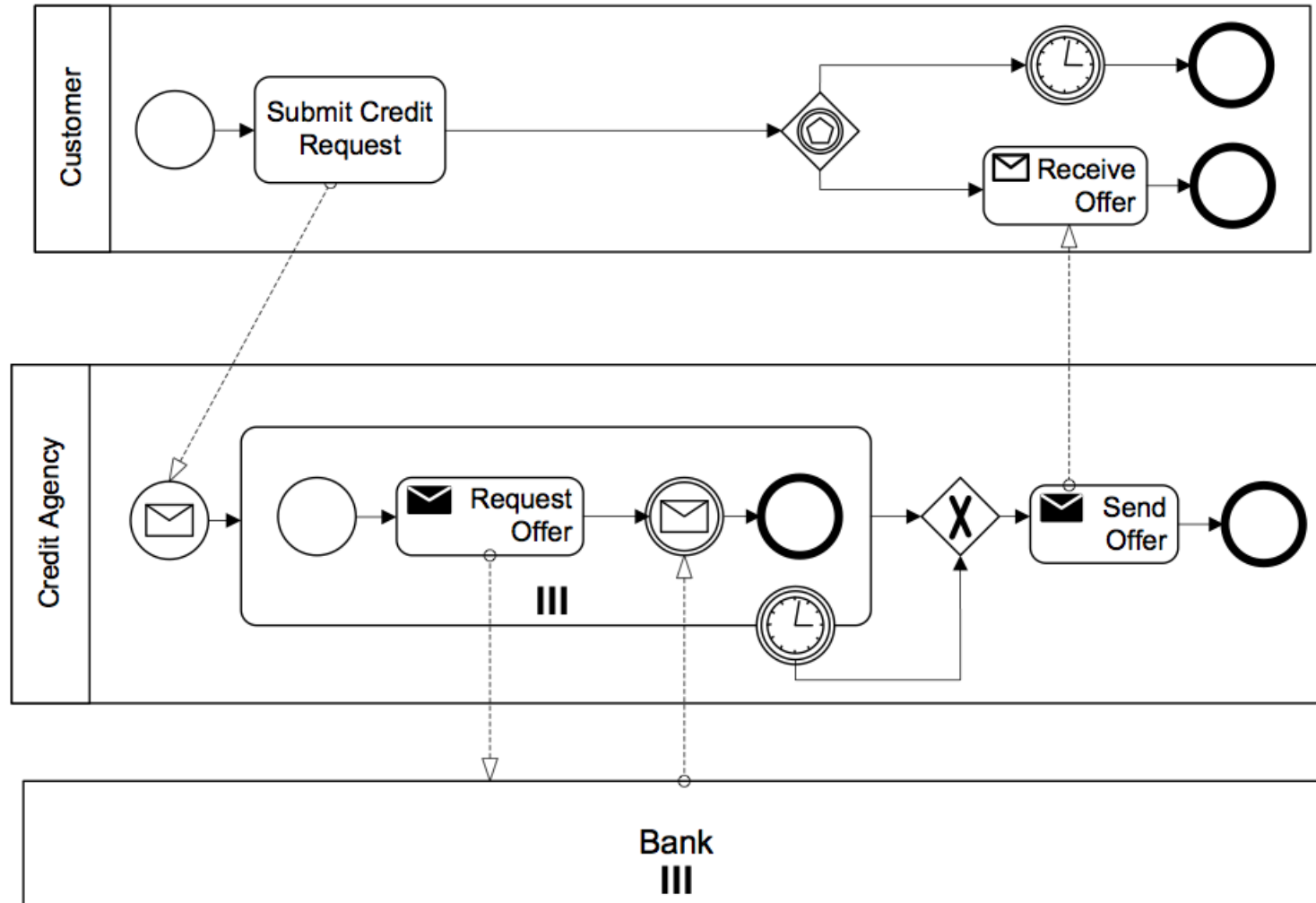
Private process



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Fig. 4.111. Collaborating business processes with private processes of both partners

Collaborating processes with a multiple instances pool



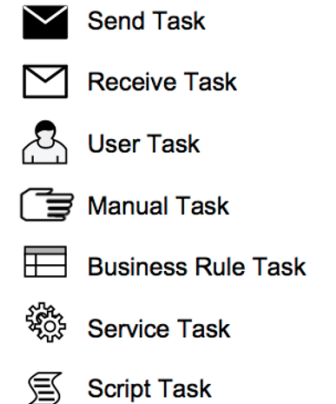
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Multi instance pool

Fig. 4.112. Collaborating processes with a multiple instances pool

Assignment: Create a Simple EPC Diagram

- A company operates a **supply chain for perishable goods** (e.g., fresh produce, vaccines...). The logistics process must handle unexpected disruptions like:
 - Delayed shipments
 - Product spoilage
 - Supplier failures
 - Regulatory compliance checks
- Model a **core BPMN process** for order fulfillment of that company
 - Introduce at least **one exception-handling pathway** (e.g., products not in stock)
 - Implement at least one **compensation or cancelation activity** (e.g., refund or product recall)
 - Also indicate the **task type** per activity



- Form a group
- Time: 30 minutes
- Objective: Familiarize yourself with a complex BPMN topic
- Result: Present your BPMN diagram to the class
- Format: Create the diagram on paper or digitally
- Help: Ask the lecturer for assistance

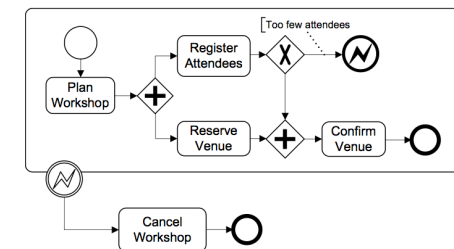


Fig. 4.92. An error is thrown in a subprocess; it is caught by an error boundary event attached to that subprocess

Questions?



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